

Supporting Information

Tailored Cis–Trans Isomeric Metal–Covalent Organic Frameworks for Coordination Configuration-Dependent Electrochemiluminescence

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Experimental Section

Materials and Reagents. Cupric acetate anhydrous, copper sulfate pentahydrate, and copper nitrate dihydrate were obtained from Shanghai Macklin Biochemical Co., Ltd. 2,4,6-trihydroxy-benzene-1,3,5-tricarbaldehyde (Tp, 98%) was purchased from Heowns Biochem. 2,3,6,7,10,11-triphenylenehexol (HHTP, 97%) and 2,3,6,7,10,11-hexaaminotriphenylene hexahydrochloride (HATP-HCl, 97%) were bought from Shanghai Bide Pharmatech Co., Ltd. 2,2,6,6-tetramethyl-4-piperidone (TEMP) and 5,5-dimethyl-1-pyrroline N-oxied (DMPO) were received from Sigma-Aldrich. Acetic acid (99.5%) and triethylamine (TEtA, 98%) were obtained from TCI Development Co., Ltd. (Shanghai). Glycine (Gly, 99%), potassium hydroxide (KOH, 99%), ammonium hydroxide (30%), ethylenediamine (en, 98%), 1,4-dioxane (Diox, 99.5%), N,N-dimethylformamide (DMF, 98%), methanol (MeOH, 99%), ethanol (EtOH, 99%), and acetone (AC, 99%) were bought from Nanjing Chemical Reagent Co., Ltd. Peroxysulphate ($K_2S_2O_8$, 98%) and L-Cysteine (Cys, 99%) were obtained from Shanghai Aladdin Bio-Chem Technology Co., Ltd. 1,3,5-trimethylbenzene (97%) was purchased from Shanghai Meryer Chemical Technology Co., Ltd. Phosphate buffer solutions (PBS, 0.10 M, pH = 7.4) were prepared by mixing stock solutions of KH_2PO_4 and K_2HPO_4 . All the reagents are used as received without further purification.

Apparatus. 1H NMR (400 MHz) and ^{13}C NMR (100 MHz) spectra were recorded on a Bruker Avance III 400 NMR spectrometer. High resolution mass spectrum (HRMS) was obtained on a Thermo Scientific Q Exactive MS. Fourier transform infrared (FTIR) spectra were taken with a spectrum on an FTIR spectrophotometer (Nicolet 6700, U.S.A.) in ATR mode at room temperature. X-ray photoelectron spectroscopy (XPS) was carried out using an ESCALAB 250 spectrometer with ultra-high vacuum generators (Thermo-VG Scientific Co., U.S.A.). Raman spectra were recorded on a laser confocal Raman microspectroscopy (Renishaw inVia-Reflex, England) at an excitation wavelength of 532 nm. Ultraviolet and visible spectra (UV-Vis) and ultraviolet diffuse reflectance spectroscopy (UV-DRS) were obtained on a UV-3600 UV-Vis-NIR spectrophotometer (Shimadzu Co., Japan). The analysis of thermal stability

was performed on a TGA/DSC 1/1100 thermal analyzer (Mettler Toledo Co., Switzerland) in the range 40-800 °C under oxygen atmosphere with a flow rate of 15 mL/min and a heating rate of 10 °C/min, and the specimen mass was about 10 mg. The energy dispersive X-ray spectroscopy mapping (EDS) images, scanning transmission electron microscopic (STEM), selected area electron diffraction (SAED) images, and transmission electron microscopic (TEM) images were recorded at an acceleration voltage of 200 kV on a JEM-2800 high resolution transmission electron microscope (JEOL Ltd., Japan). Scanning electron microscopy (SEM) images were recorded using a JEOL JSM-7800F field emission scanning electron microscope. N₂ adsorption-desorption isotherms were characterized via the Brunauer-Emmett-Teller (BET) method on a JW-BK200B volumetric gas sorption instrument at 77 K. Powder X-ray diffraction (PXRD) data were collected on a Bruker D8-Advance diffractometer with a Cu-sealed tube ($\lambda = 1.54178$ Å) at 40 kV and 40 mA with 0.1 s per step. Crystal models were generated using Materials Studio version 2019 software. Electron paramagnetic resonance (EPR) spectral measurements were performed at room temperature on Bruker Elexsys E580 X-band. Electrochemical impedance spectroscopy (EIS) was gained from a Parstat MC 1000 electrochemical workstation (Princeton, USA) in 0.1 M KCl solution containing 5 mM K₃Fe(CN)₆/K₄Fe(CN)₆ (1:1).

ECL Measurement. The ECL measurements were conducted on MPI-E II multifunctional electrochemical and chemiluminescent analytical system (Xi'an Remex Analytical Instrument Co., Ltd. China) in a conventional three-electrode cell in the potential range from 0.0 V to -1.7 V with a scan rate of 0.1 V s⁻¹ at a 700 V photomultiplier tube (PMT). The three electrodes include a 5-mm diameter glassy carbon electrode (GCE) as working electrode, a Pt counter electrode, and a Ag/AgCl (saturated) reference electrode. By default, the samples were added into ultrapure water to form 1.0 mg mL⁻¹ dispersion with sonicating for 10 min. 20 µL of dispersion was dropped on the GCE surface, and then dried at room temperature under an ambient atmosphere. The ECL measurements were conducted in 0.1 M pH 7.4 PBS

containing 0.1 M KCl as the electrolyte and 0.1 M K₂S₂O₈ as the coreactant using cyclic voltammetry (CV).

ECL Efficiency Calculation. The relative ECL efficiencies were evaluated by the ratios of Φ_{ECL} , which can be calculated as below¹:

$$\Phi_{ECL} = \frac{\int_0^t I dt}{\int_0^{t'} i_a dt} = \frac{\int_0^{t'} I dt}{Q_a}$$

Where I (einsteins s⁻¹) represents the total ECL intensity integrated over the testing period of time t' (s) and i_a represents integration of the anodic current over the same time period that results in the total anodic charge Q_a .

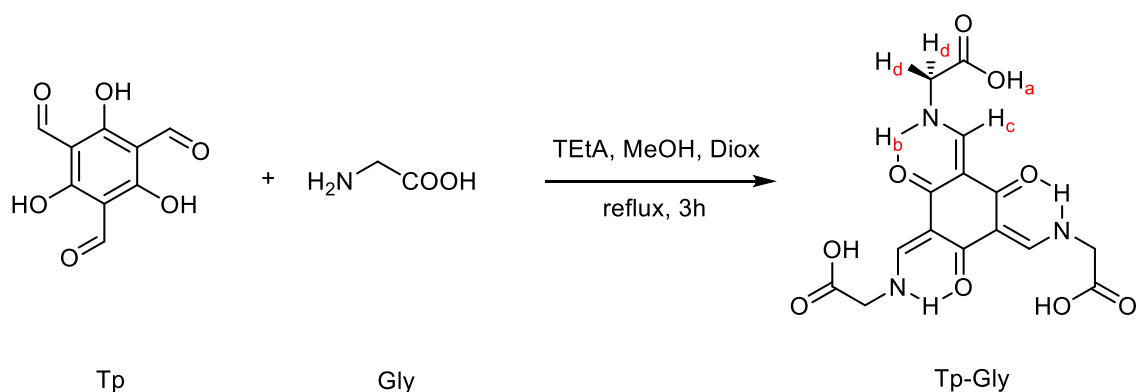
ECL Spectrum. ECL spectrum was recorded on a homemade ECL spectrum analyzer consisting of a CHI-660D electrochemical workstation equipped with an Edinburgh FLS980 Series of Fluorescence Spectrometers.

Synthesis of Cis-bis(glycinato)copper(II) (E-Cu(Gly)₂). Weigh basic copper carbonate (2.0 g, 9.05 mmol) and Gly (3.0 g, 40 mmol) into a 250 mL beaker. Add 100 mL of deionized water to the mixture and allow the solution to sit until the cessation of gas production. Initiate heating while ensuring the temperature does not exceed 65 °C. Cease heating once half of the initial volume of the solution in the beaker has evaporated. Following complete cooling of the solution, proceed to pump and filter it. Conduct three rounds of washing using small quantities of deionized water. Subsequently, dry the resulting mixture with an infrared lamp to acquire sky-blue, needle-like powder. Yield: 0.48 g, 25%.

Synthesis of Trans-bis(glycinato)copper(II) (Z-Cu(Gly)₂). Basic copper carbonate (2.0 g, 9.05 mmol) was placed in a 250 mL beaker containing 75 mL of deionized water, and gradually added Gly (4.0 g, 53.3 mmol) in batches. Heat the mixture to 85 °C. Wait until no gas is being produced, and then discontinue the heating process. Observe the gradual formation of dark blue solid precipitation. Once the

temperature has fallen below 50 °C, introduce 75 mL of deionized water and continue stirring. Notice the progressive manifestation of a metallic luster within the solid. After the solid has completely settled, proceed with pumping and filtering the mixture. Perform three rounds of washing using small quantities of deionized water. Subsequently, the solid was dried under an infrared lamp to yield a dark blue scaly powder. Yield: 0.80 g, 42%.

Synthesis of Tp-Gly. Tp (90 mg, 0.43 mmol), Gly (193 mg, 2.6 mmol), and KOH (146 mg, 2.6 mmol) are accurately weighed and placed into a 100 ml round-bottom flask. Subsequently, a mixture comprising Diox (20 mL) and MeOH (40 mL) is added to the flask. Then 300 μ L of TEtA is carefully introduced into the reaction mixture. The ensuing chemical transformation is carried out through a reflux process at a controlled temperature of 85 °C for 3 h. Following the completion of the reflux, separation of insoluble constituents is achieved by employing a filtration process. Furthermore, 15 mL of 3 M HCl solution was introduced into the system. After the reaction is complete, the resultant mixture is subjected to vacuum filtration to isolate the ensuing precipitate. To ensure thorough purification, the filter cake is subjected to sequential individual washings using 5 mL portions of H₂O, EtOH, and AC, each repeated three times. The final step involves the drying of the obtained Tp-Gly ligand under the infrared light. Yield: 154 mg, 94%. ¹H NMR (DMSO-d₆ 400MHz): δ = 12.98 (br, 3H_a), 11.23-10.80 (m, 3H_b), 8.23-8.02 (m, 3H_c), 4.30 (t, 6H_d). ¹³C NMR (DMSO-d₆, 100 MHz): δ = 184.59, 171.28, 158.09, 104.97, 50.20. HRMS (m/z) [M-H]⁻, [C₁₅H₁₅N₃O₉-H]⁻: Calc. 380.0735, found 380.0963.



Synthesis of E-MCOF. In a 5 mL scintillation vial, Tp-Gly (5.10 mg, 0.0134 mmol) and copper nitrate dihydrate (3.76 mg, 0.02 mmol) were mixed with DMF (1.0 mL) and MeOH (1.0 mL). After ultrasonic mixing, introduce 2.8 μ L (0.02 mmol) of Et₃N and shake the mixture well. Keep the vial at 37 °C for one week, followed by a day-long heating at 100 °C. Collect the product by centrifugation, wash with 3 mL H₂O, EtOH and AC, respectively. Finally, the sample was placed in a vacuum oven at 80 °C for 12 h to obtain the E-MCOF powder.

Synthesis of Z-MCOF. Weigh Tp (25.22 mg, 0.12 mmol) and Z-Cu(Gly)₂ (38.10 mg, 0.18 mmol) into a mortar, grind to obtain a homogeneous mixture. Add 50 μ L of triethylamine to the powdered mixture, and grind rapidly until a viscous sample is formed. After excess triethylamine is removed using an infrared lamp, add 50 μ L of water and grind until a slurry sample is formed. Transfer the mixture to a preheated oven at 100 °C for 15 min. After cooling, detach the product from the mortar wall, denoting it as the precursor. Take 50 mg of the precursor and place it into a 15 mL glass tube, add 2 mL of dioxane and 10 μ L of water, sonicate for uniform dispersion. The tube was degassed by the freeze–pump–thaw technique three times to achieve an internal pressure of ~50 mTorr and flame sealed under vacuum. The reaction was heated at 90 °C for 72 h to yield green solid at the bottom of the tube. Centrifuge and separate the sample, perform solvent exchange with 20 mL MeOH and AC, respectively. Finally, the sample was placed in a vacuum oven at 80 °C for 12 h to obtain the Z-MCOF powder. Z-MCOF powder was dispersed in water and coated dropwise on a slide to form a film for PXRD testing².

Synthesis of Z-MCOF₀ (0 Degree Twisted). Tp (25.22 mg, 0.12 mmol) and Z-Cu(Gly)₂ (38.10 mg, 0.18 mmol) are weighed into a 15 mL glass tube. 2 mL of mesitylene and 100 μ L of triethylamine were added to the tube, which was sonicated thoroughly. The tube was degassed by the freeze–pump–thaw technique three times to achieve an internal pressure of ~50 mTorr and flame sealed under vacuum. The reaction was heated at 90 °C for 72 h to yield dark brown solid at the bottom of the tube. Centrifuge and separate

the sample, perform solvent exchange with 20 mL MeOH and AC, respectively. Finally, the sample was placed in a vacuum oven at 80 °C for 12 h to obtain a dark brown powder.

Synthesis of Cu-HITP. The synthesis method was derived from the published literature³. 5 mL of deionized water was added into a 20 mL scintillation vial, following with CuSO₄ (2.458 mg, 0.0154 mmol), and 300 µL of 30 % ammonium hydroxide while 10 mL of water and HATP-HCl (40.526 mg, 0.077 mmol) were added into a separate 20 mL vial. The second vial was poured into the first vial and then heated at 80 °C for 24 h exposed to air. After finished, the reaction volume was filtered and washed with 30 mL of deionized water and acetone, respectively. Finally, the sample was placed in a vacuum oven at 50 °C for 12 h to obtain a dark purple powder.

Synthesis of Cu-HHTP. The synthesis method was adapted from the published literature³. To a 20 mL scintillation vial, HHTP (19.46 mg, 0.06 mmol) and cupric acetate anhydrous (32.69 mg, 0.18 mmol) were added. 5 mL of deionized water was added to the scintillation vial. The solution was sonicated for 10 min. The reaction mixture was heated at 85 °C for 24 h. The product was then filtered through a ceramic funnel and filter paper and washed with 30 mL of deionized water and acetone, respectively. Finally, the sample was placed in a vacuum oven at 50 °C for 12 h to obtain a black powder.

Calculation Details

The first-principles calculations were performed using a projected augmented wave (PAW) method⁴, as implemented in the Vienna ab initio simulation package (VASP)⁵⁻⁶. The Van de Waals (vdW) correction and spin polarization were taken into consideration. Bader charges were calculated by taking superposition of atomic charge densities. Geometry optimizations were performed by using the generalized gradient approximation (GGA) and the Perdew-Burke-Ernzerhof (PBE)⁷. The adsorption energies were calculated on models of Z-MCOF bulk, Z-MCOF surface and E-MCOF surface, respectively. The vacuum layer was set to be 20 Å on the surface models of Z-MCOF and E-MCOF. The

convergence for force was $0.04 \text{ eV} \cdot \text{\AA}$ and $1 \times 10^{-5} \text{ eV}$ for the total energy. The cut-off kinetic energy was set to be 450 eV, and gamma-point-centered k-point density was $0.04 \text{ \AA}^{-1}$. The Heyd-Scuseria-Ernzerhof (HSE06)⁸ hybrid functional was used to calculate the density of states (DOS) for Cu-HHTP, Cu-HITP, E-MCOF and Z-MCOF materials. Taking into account the amount of computation, the parameter settings for the DOS calculations were as follows: (1) the cut-off energy of 500 eV, (2) an electronic convergence criterion of 10^{-5} eV . In addition, the adsorption modes of $\text{K}_2\text{S}_2\text{O}_8$ on E-MCOF were explored using the COMPASS force field in Materials Studio.

In our calculations, the adsorption energy (E_{ads}) was defined as follows:

$$E_{\text{ads}} = E(\text{K}_2\text{S}_2\text{O}_8@\text{cata}) - E(\text{cata}) - E(\text{K}_2\text{S}_2\text{O}_8)$$

Where $E(\text{K}_2\text{S}_2\text{O}_8@\text{cata})$ was the total energy of $\text{K}_2\text{S}_2\text{O}_8$ absorbed in Z-MCOF surface, $E(\text{cata})$ was the energy of Z-MCOF, and $E(\text{K}_2\text{S}_2\text{O}_8)$ was the energy of $\text{K}_2\text{S}_2\text{O}_8$.

Supporting Figures

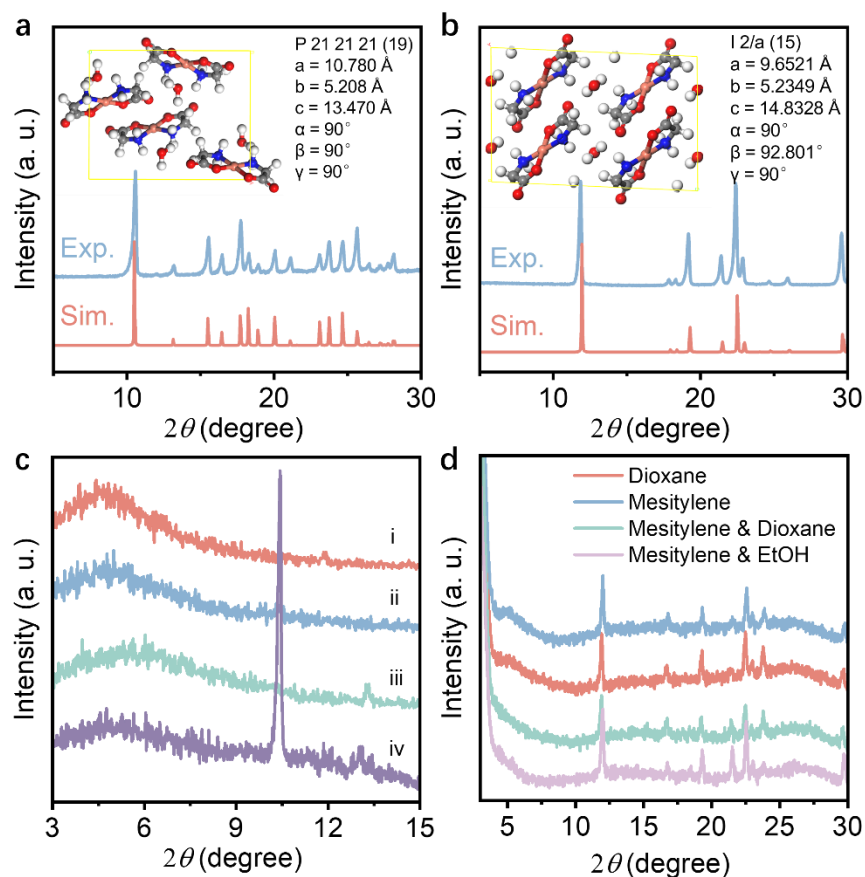


Figure S1. Experimentally obtained (blue) and simulated from single-crystal (red) PXRD patterns of (a) E-Cu(Gly)₂ and (b) Z-Cu(Gly)₂. Inset: single crystal cell and lattice parameters. Carbon, hydrogen, oxygen, nitrogen and copper atoms are represented as gray, white, red, blue and orange spheres, respectively. (c) PXRD patterns of the productions via solvent-thermal reaction approach for Tp + Z-Cu(Gly)₂ in mesitylene (i), Tp + E-Cu(Gly)₂ in mesitylene (ii), Tp + Z-Cu(Gly)₂ in dioxane (iii), and Tp + E-Cu(Gly)₂ in dioxane (iv). (d) PXRD patterns of Z-MCOF synthesized by solvothermal method in different solvents.

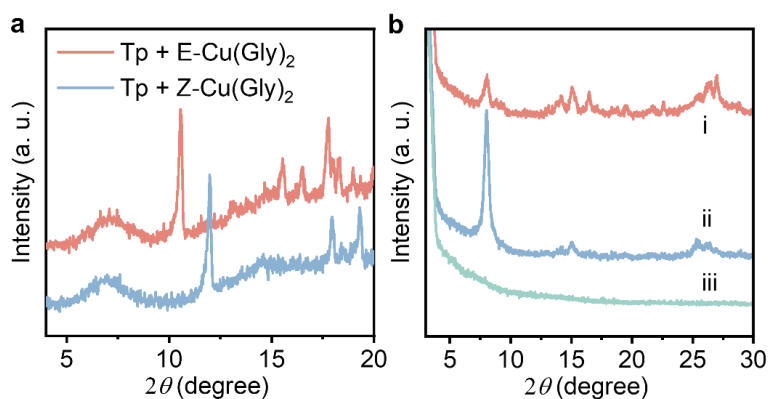


Figure S2. PXRD patterns of (a) MCOF precursors synthesized by solvent-assisted grinding for solid Tp + E-Cu(Gly)₂, and Tp + Z-Cu(Gly)₂. (b) Optimization of TETA content at 0 (i), 0.02 mmol (ii), and 0.04 mmol (iii) in the kinetics dominated synthesis of E-MCOF.

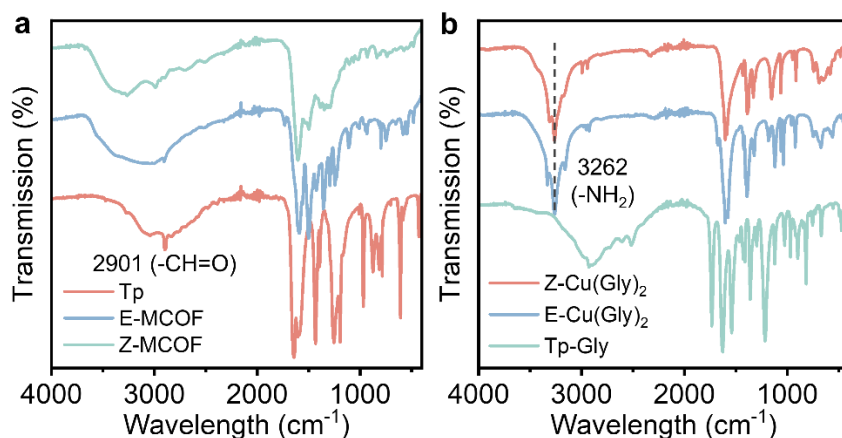


Figure S3. Stacked FT-IR spectra of the reaction substrates in the synthesis of MCOFs.

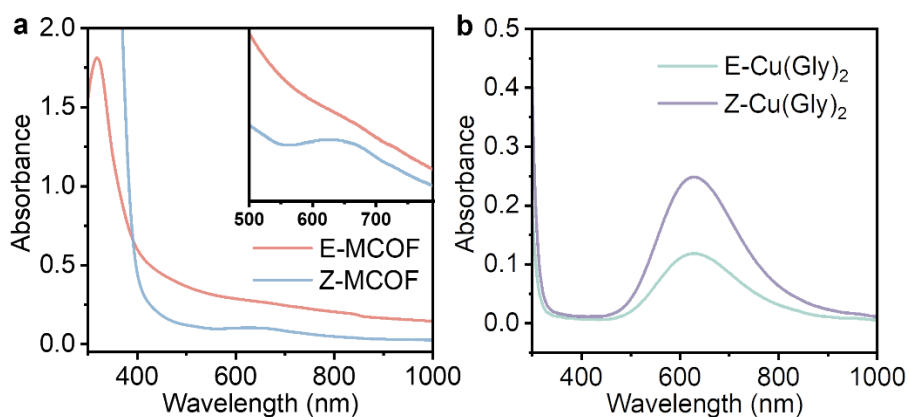


Figure S4. UV-vis spectra of (a) MCOFs and (b) Cu(Gly)₂. Inset of (a): zoom of 500 – 800 nm region.

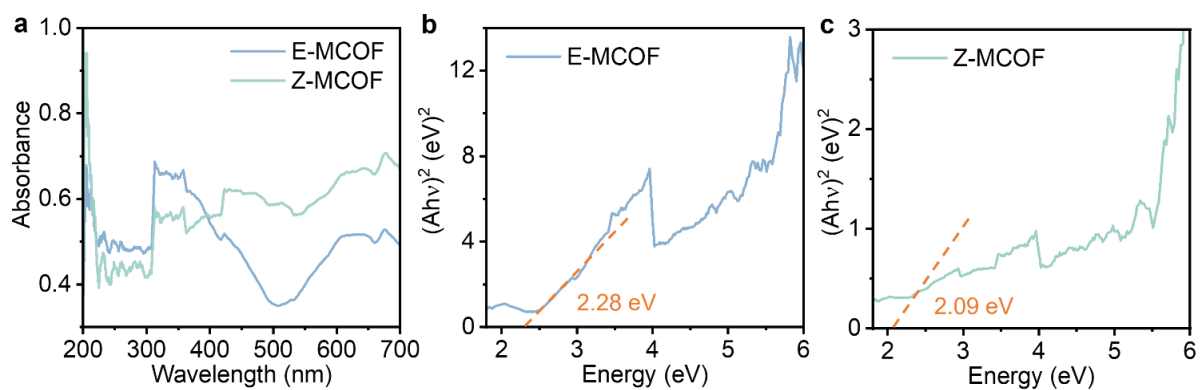


Figure S5. (a) UV-DRS spectra of MCOFs. Corresponding Tauc plots of (b) E-MCOF and (c) Z-MCOF.

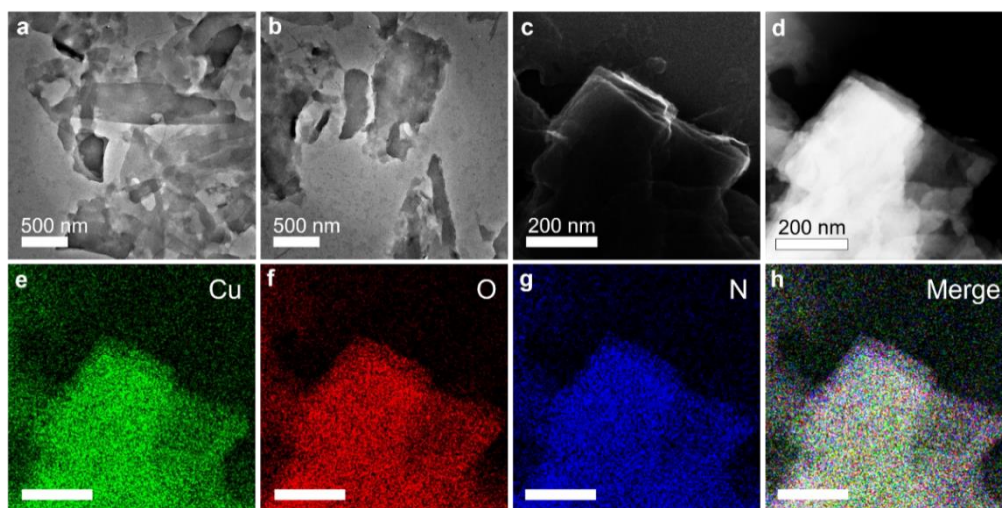


Figure S6. TEM (a, b), STEM (c, d) and EDS mapping (e-h) images of E-MCOF.

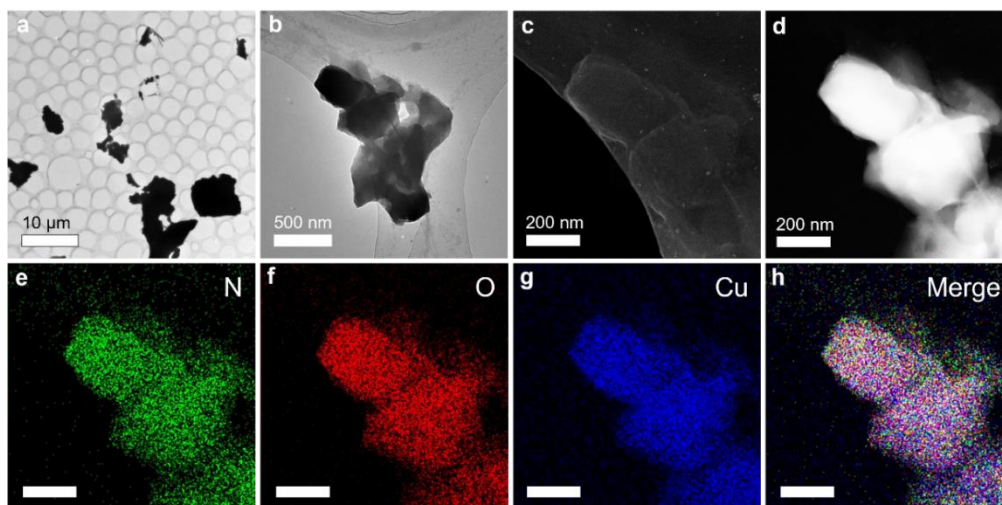


Figure S7. TEM (a, b), STEM (c, d) and EDS mapping (e-h) images of Z-MCOF.

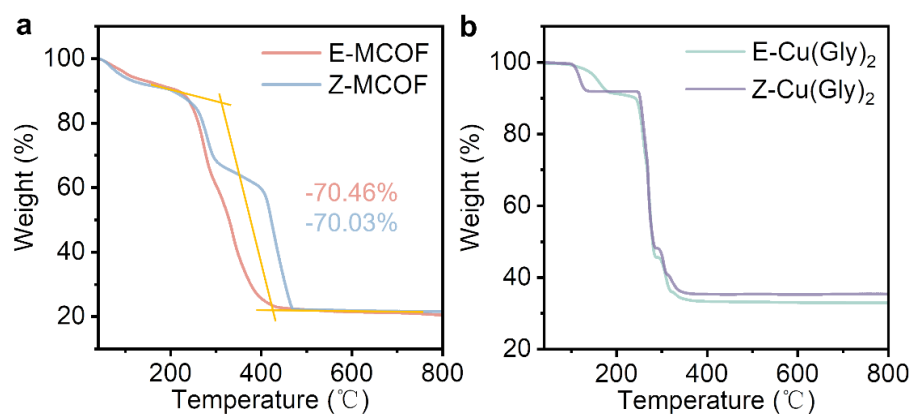


Figure S8. TGA of (a) MCOFs and (b) Cu(Gly)₂ collected under air atmosphere.

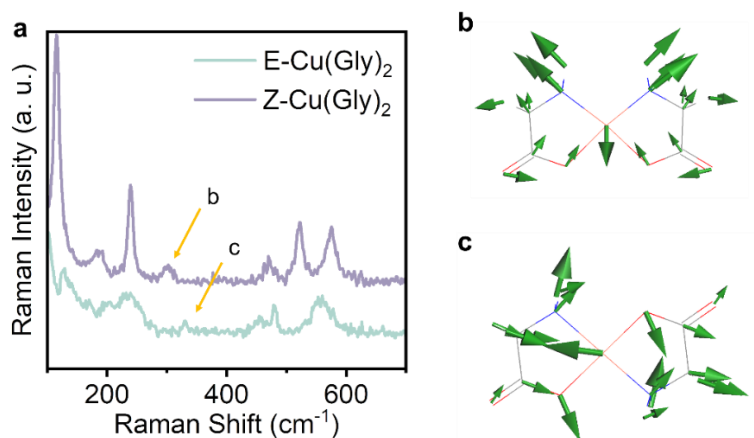


Figure S9. (a) Raman spectra of Cu(Gly)₂ at 298 K with 532 nm excitation. The Raman characteristic vibration mode simulated by the Dmol3 module of the Materials Studio software of (b) E-Cu(Gly)₂ at 330.54 cm⁻¹ and (c) Z-Cu(Gly)₂ at 307.77 cm⁻¹. The arrows in the image indicate the direction of motion of the atoms located at their ends in the vibration and the sizes of the arrows indicate the magnitudes of the atomic displacement.

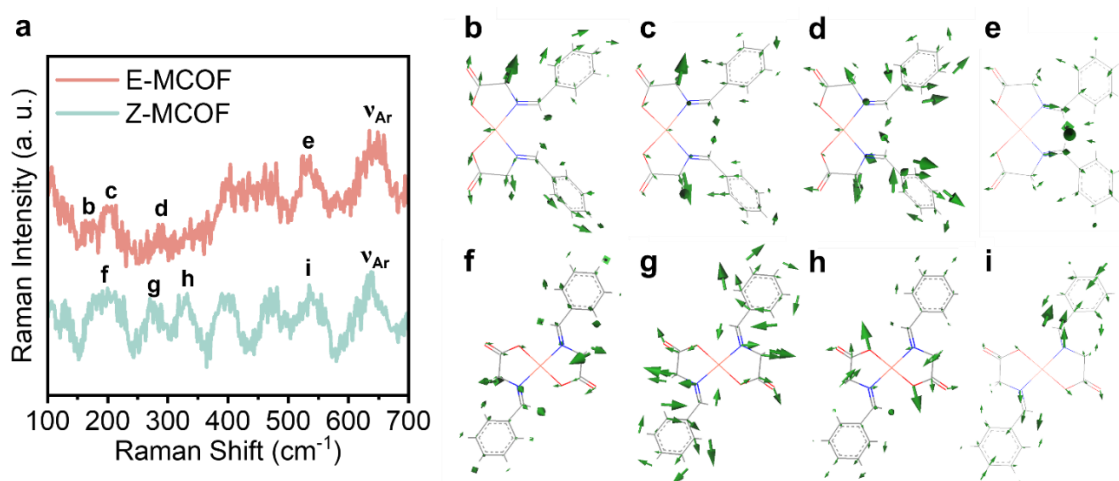


Figure S10. (a) Raman spectra of MCOFs at 298 K with 532 nm excitation. The Raman characteristic vibration mode simulated by the Dmol3 module of the Materials Studio software of E-MCOF fragment at (b) 161.26 cm^{-1} , (c) 207.59 cm^{-1} , (d) 278.84 cm^{-1} and (e) 542.81 cm^{-1} , and Z-MCOF fragment at (f) 196.09 cm^{-1} , (g) 289.36 cm^{-1} , (h) 324.66 cm^{-1} and (i) 547.54 cm^{-1} . The arrows in the images indicate the direction of motion of the atoms located at their ends in the vibration and the sizes of the arrows indicate the magnitudes of the atomic displacement.

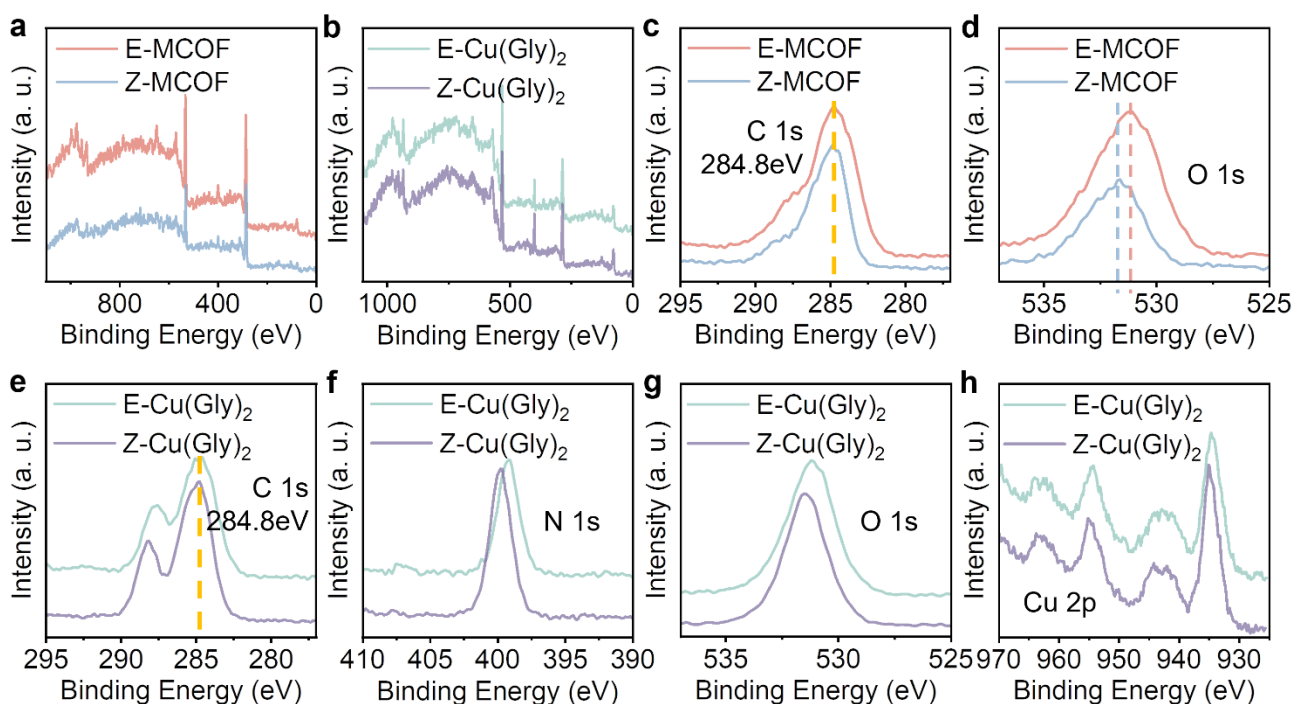


Figure S11. XPS survey spectra of (a) MCOFs and (b) $\text{Cu}(\text{Gly})_2$. XPS spectra of MCOFs in (c) C 1s and (d) O 1s regions. XPS spectra of $\text{Cu}(\text{Gly})_2$ in (e) C 1s, (f) N 1s, (g) O 1s, and (h) Cu 2p regions.

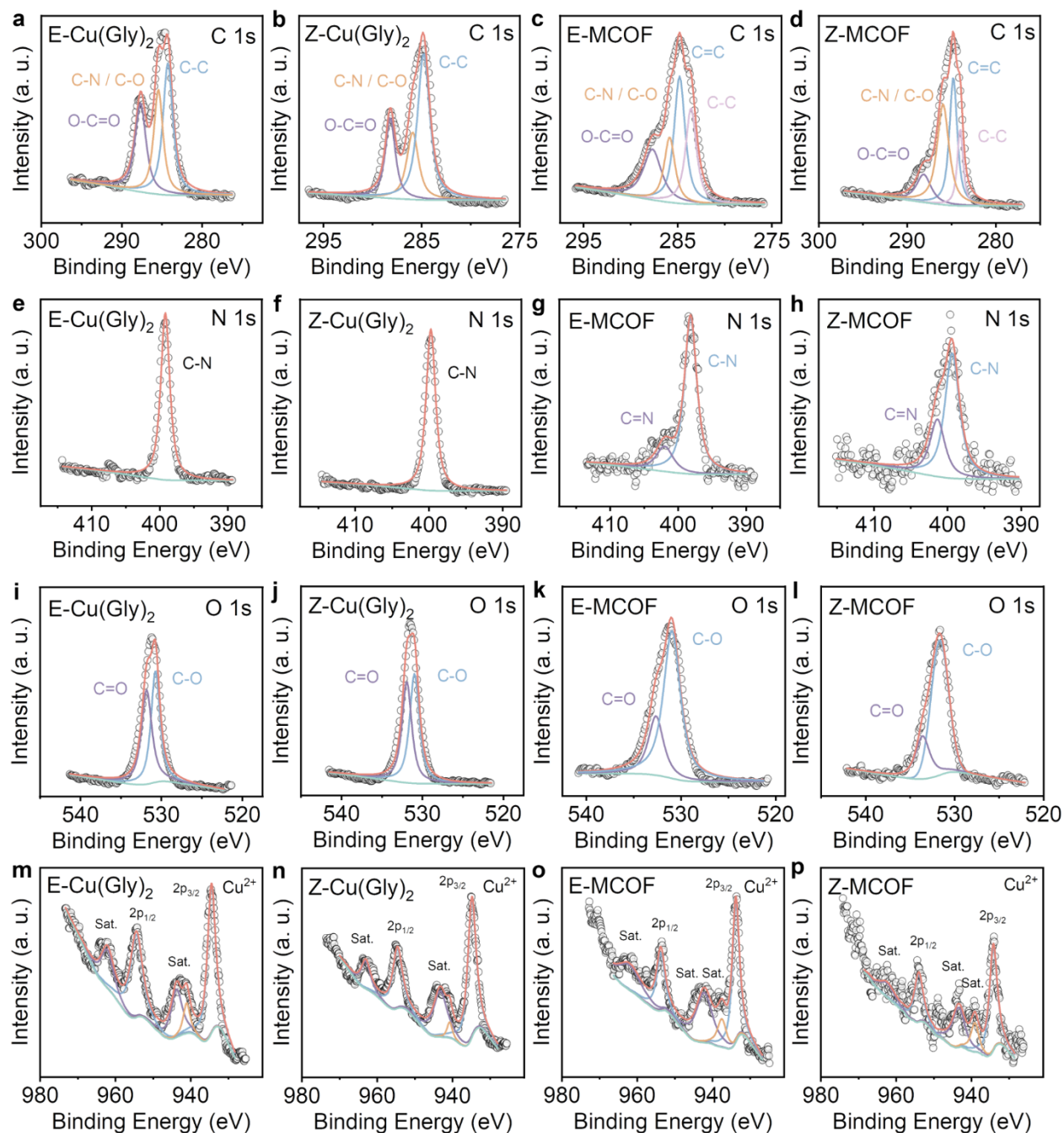


Figure S12. The deconvolutions of XPS peaks of (a-d) C, (e-h) N, (i-l) O, and (m-p) Cu elements of MCOFs and Cu(Gly)₂.

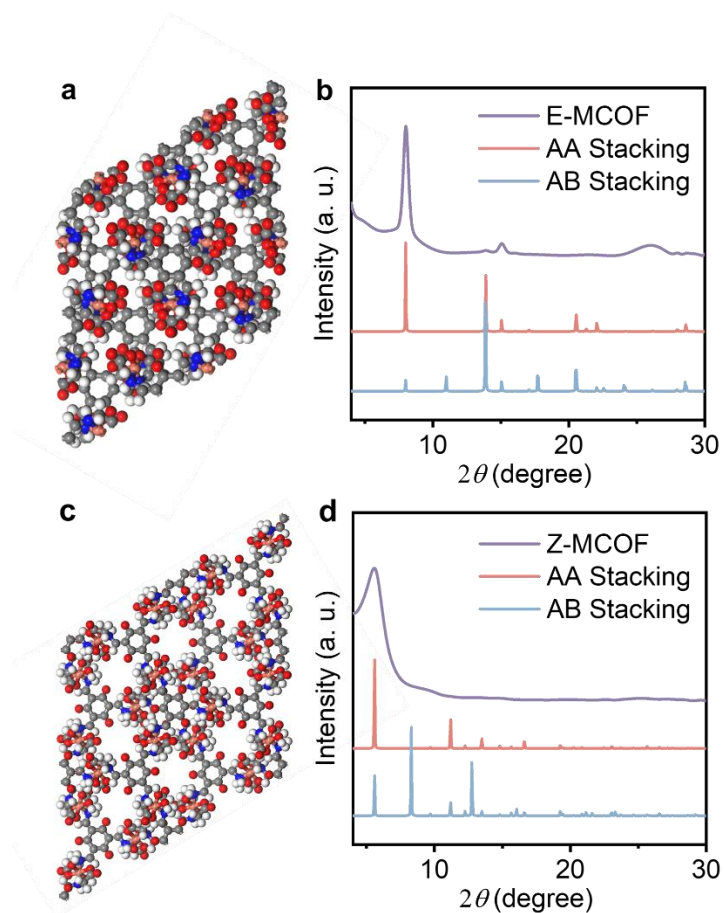


Figure S13. AB stacking modes and the simulated PXRD patterns of (a, b) E-MCOF and (c, d) Z-MCOF.

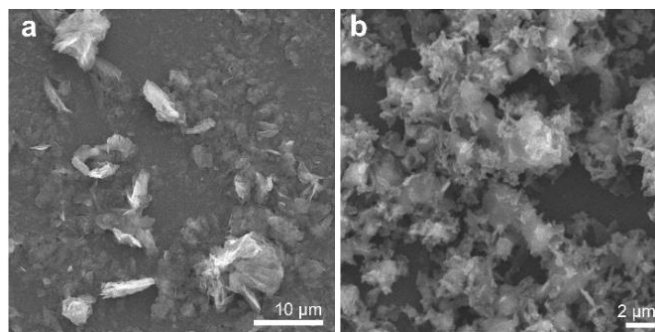


Figure S14. SEM images of E-MCOF dispersed in the aqueous phase at different scale bars: (a) 10 μm and (b) 2 μm .

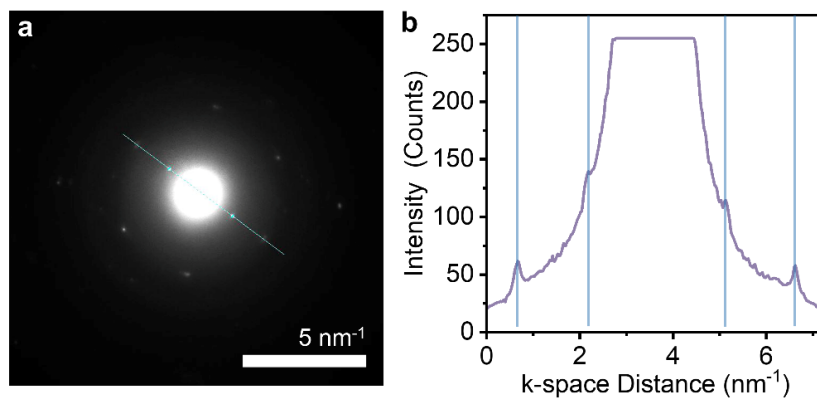


Figure S15. (a) SAED pattern of E-MCOF. (b) Pixel brightness distribution map along the line in a.

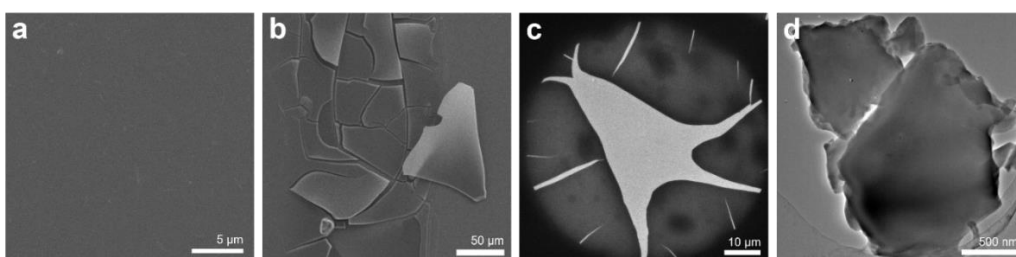


Figure S16. SEM (a, b) and TEM (c, d) images of Z-MCOF dispersed in the aqueous phase.

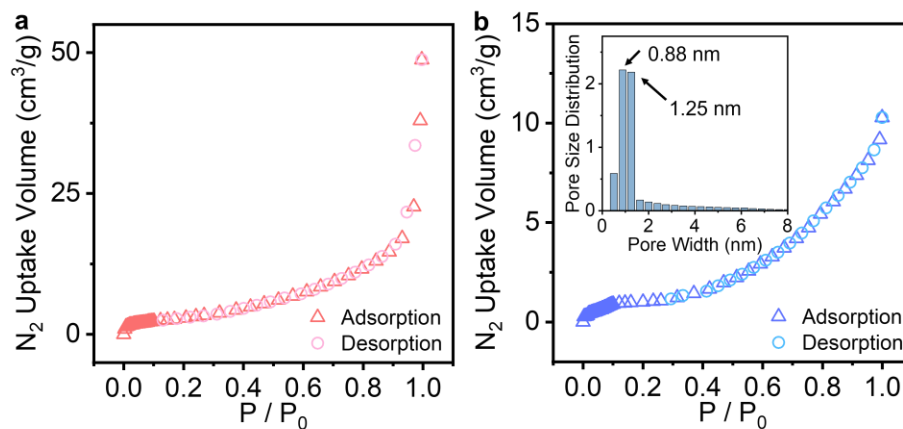


Figure S17. N₂ isotherms of (a) E-MCOF and (b) Z-MCOF at 77 K. Insets: pore-size distribution fitted by NLDFT.

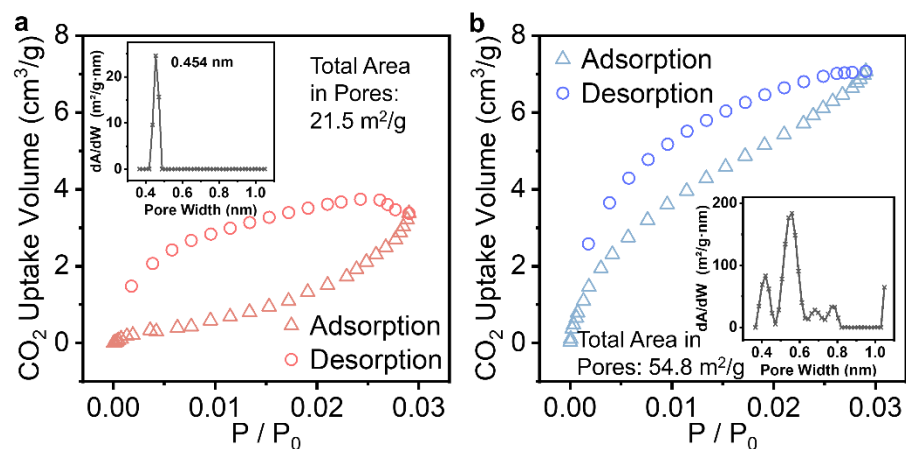


Figure S18. CO₂ isotherms of (a) E-MCOF and (b) Z-MCOF at 273 K. Insets: pore-size distribution fitted by NLDFT.

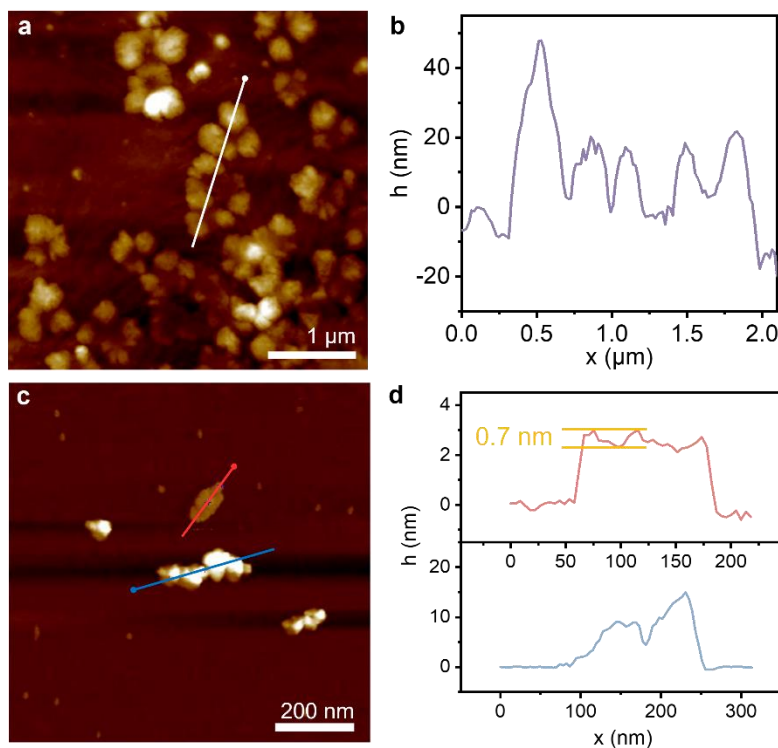


Figure S19. (a, c) AFM images of the aggregate states of Z-MCOF. (b, d) Height distribution of the underlined portions.

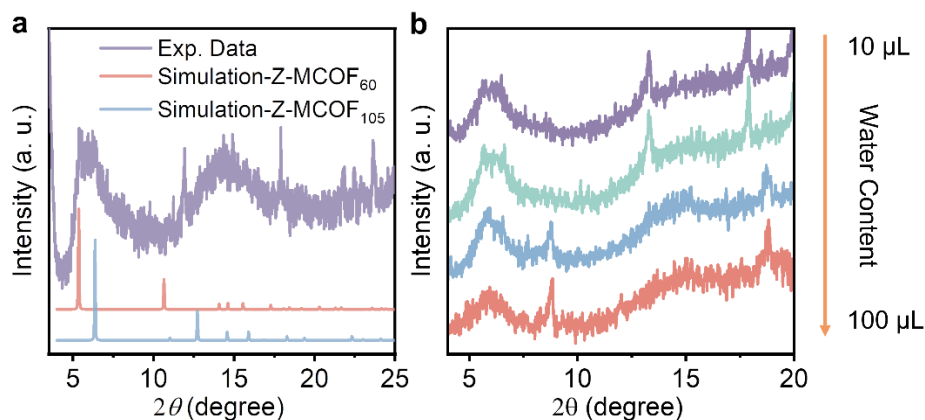


Figure S20. (a) PXRD patterns of the unregulated Z-MCOF and simulated PXRD patterns of Z-MCOF with a 60° (red) and 105° (blue) twist angles. (b) PXRD patterns of Z-MCOF-precursor regulated with different water contents in 2 mL dioxane: 10 μ L (purple), 20 μ L (green), 50 μ L (blue), and 100 μ L (red).

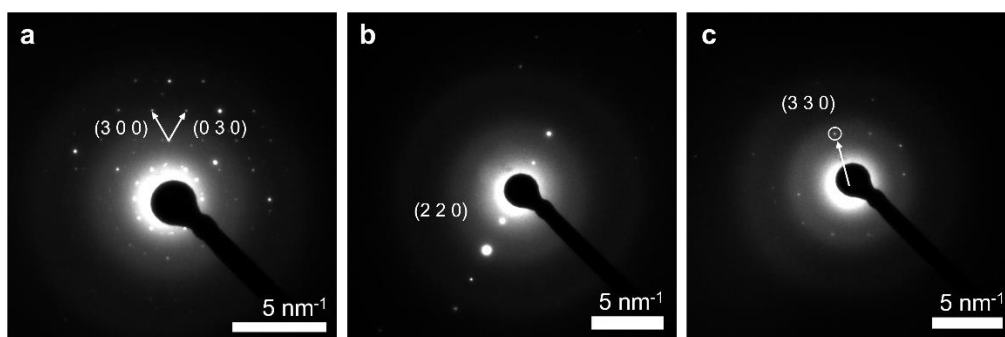


Figure S21. SAED patterns of (a) Z-MCOF₆₀, and (b,c) Z-MCOF₁₀₅.

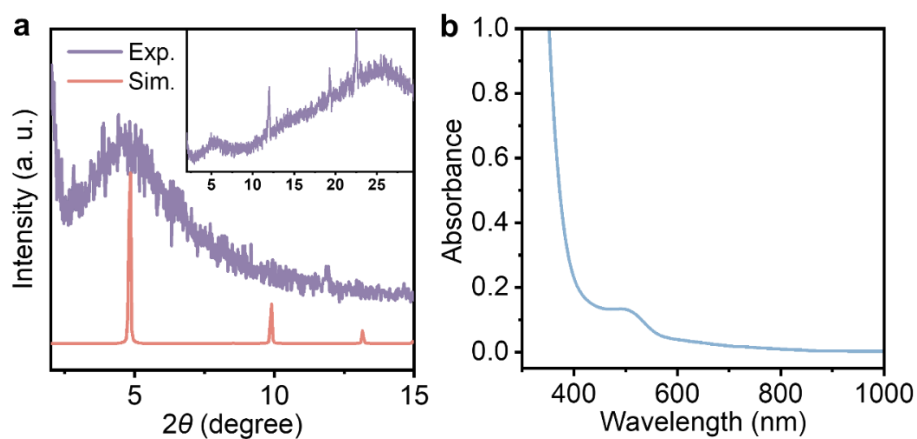


Figure S22. (a) The simulated and observed PXRD patterns of Z-MCOF₀. Inset: wide-angle PXRD pattern. (b) UV-vis spectrum of Z-MCOF₀.

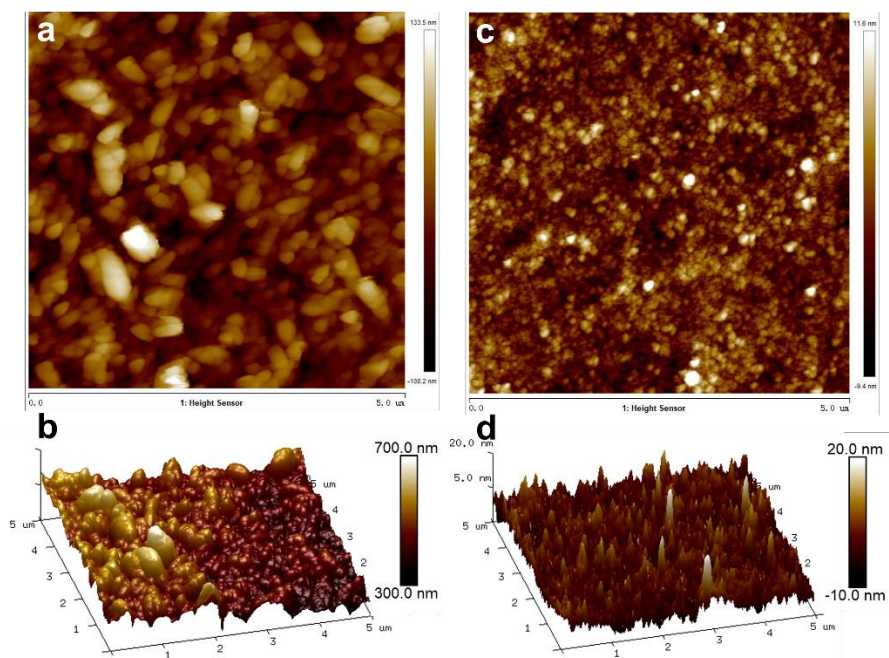


Figure S23. Two-dimensional and three-dimensional AFM images of (a, b) E-MCOF and (c, d) Z-MCOF.

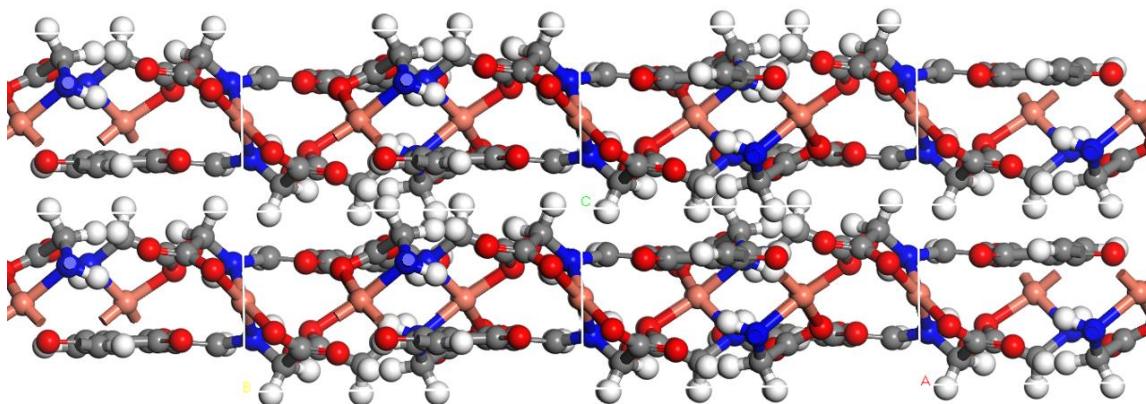


Figure S24. Enlarged side view of E-MCOF structure. Carbon, hydrogen, oxygen, nitrogen and copper atoms are represented as gray, white, red, blue and orange spheres, respectively.

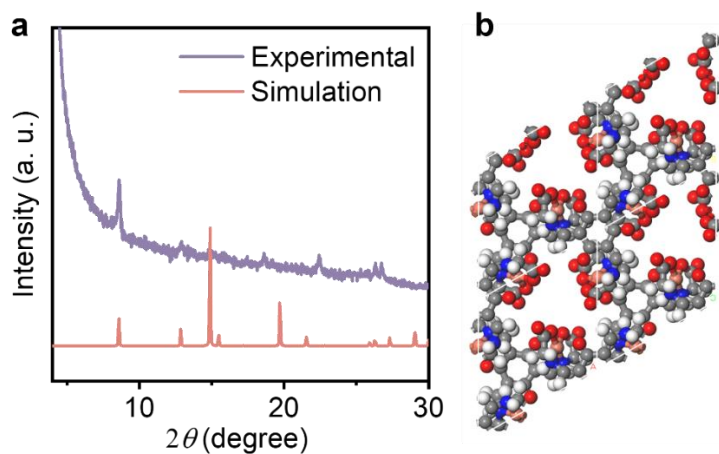


Figure S25. (a) The simulated and observed PXRD patterns of E-MCOF heated at 90 °C for 12 h. (b) Top view of simulated crystal structure in (a).

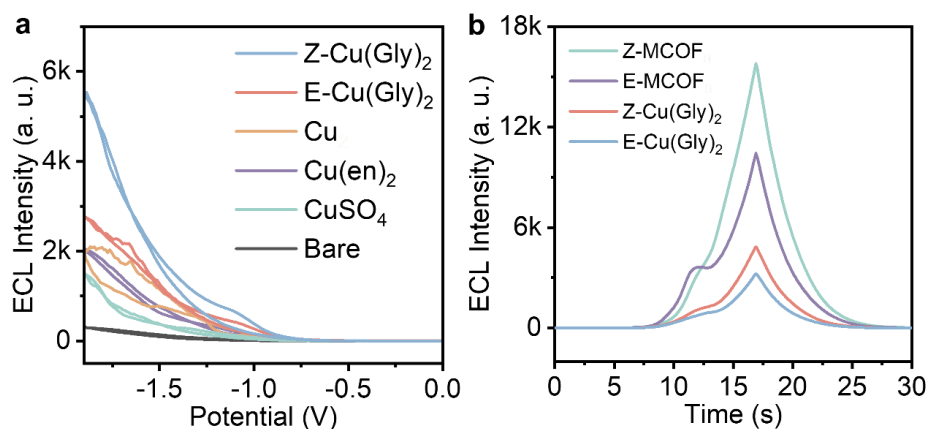


Figure S26. (a) ECL curves of different copper complexes dispersed in dioxane modified GCEs from 0 V to -1.9 V in 0.1 M PBS (pH = 7.4) containing 0.1 M $K_2S_2O_8$ (PMT = 600 V). (b) ECL curves of 1 mg/mL MCOFs and 0.67 mg/mL $Cu(Gly)_2$ dispersed in dioxane modified GCEs from 0 V to -1.7 V in 0.1 M PBS (pH = 7.4) containing 0.1 M $K_2S_2O_8$ (PMT = 700 V). Scan rate: 0.1 V s⁻¹.

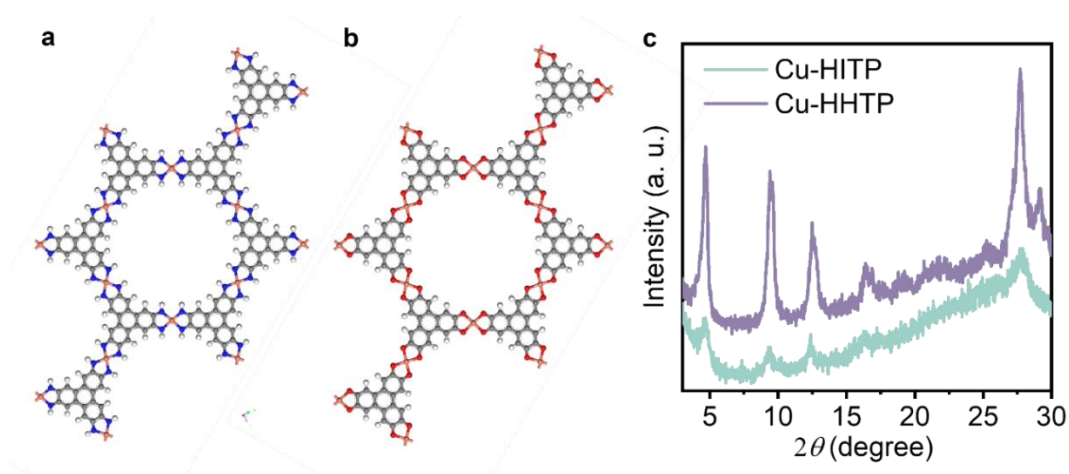


Figure S27. Simulated crystal structures of (a) Cu-HITP and (b) Cu-HHTP. (c) PXRD patterns of Cu-HITP and Cu-HHTP.

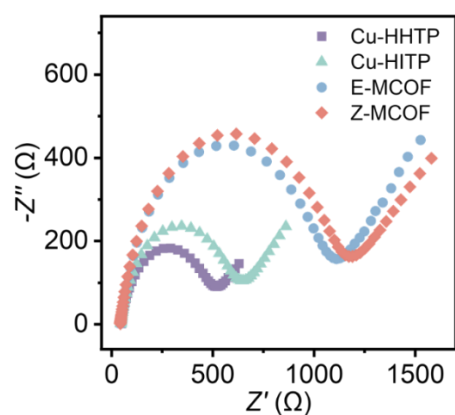


Figure S28. EIS of four frameworks modified GCEs in 0.1 M KCl containing 5.0 mM $[\text{Fe}(\text{CN})_6]^{3/4-}$.

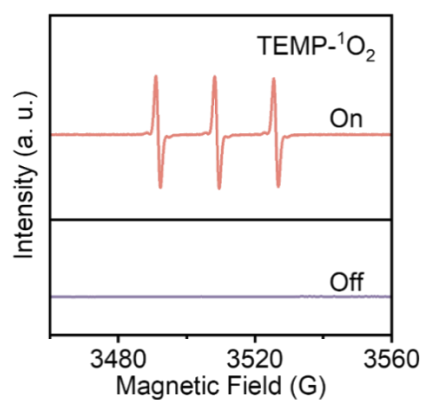


Figure S29. TEMP-trapped EPR spectra of luminescent species in ECL process of Z-MCOF before (purple) and after (red) applying potentials.

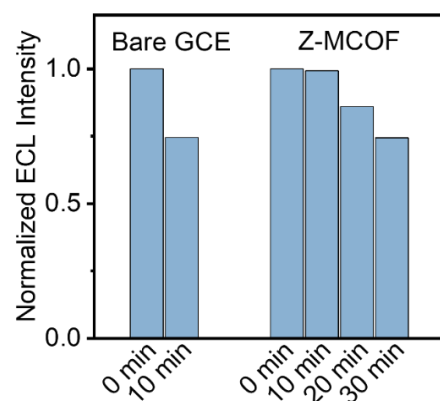


Figure S30. Deoxygenation ECL tests of bare GCE and Z-MCOF modified electrodes for different nitrogen bubbling time.

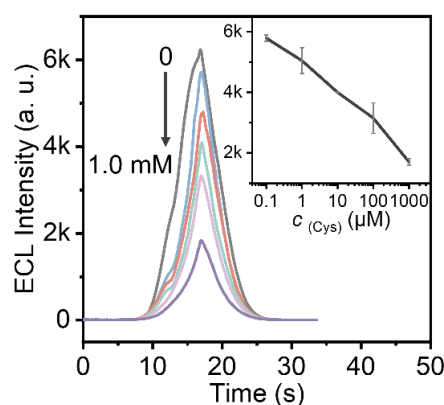


Figure S31. ECL response of Z-MCOF to different concentrations of L-cysteine dispersed in 0.1 M pH 7.4 PBS containing 0.1 M $\text{K}_2\text{S}_2\text{O}_8$. Concentrations of L-cysteine: 0, 0.1, 1.0, 10, 100, 1000 μM (from up to bottom). Inset: correlation between L-cysteine concentration and ECL intensity.

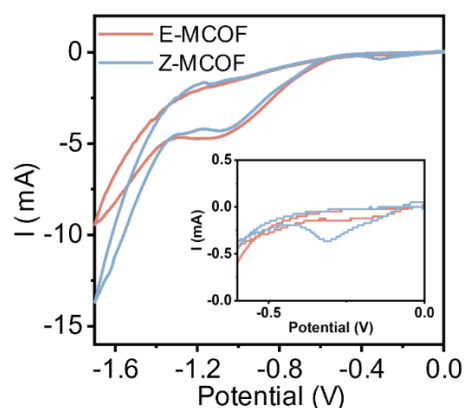


Figure S32. CV curves of MCOFs modified GCEs in 0.1 M PBS (pH = 7.4) containing 0.1 M $\text{K}_2\text{S}_2\text{O}_8$. Scan rate: 0.1 V s^{-1} . Inset: enlarged view from 0 V to -0.6 V.

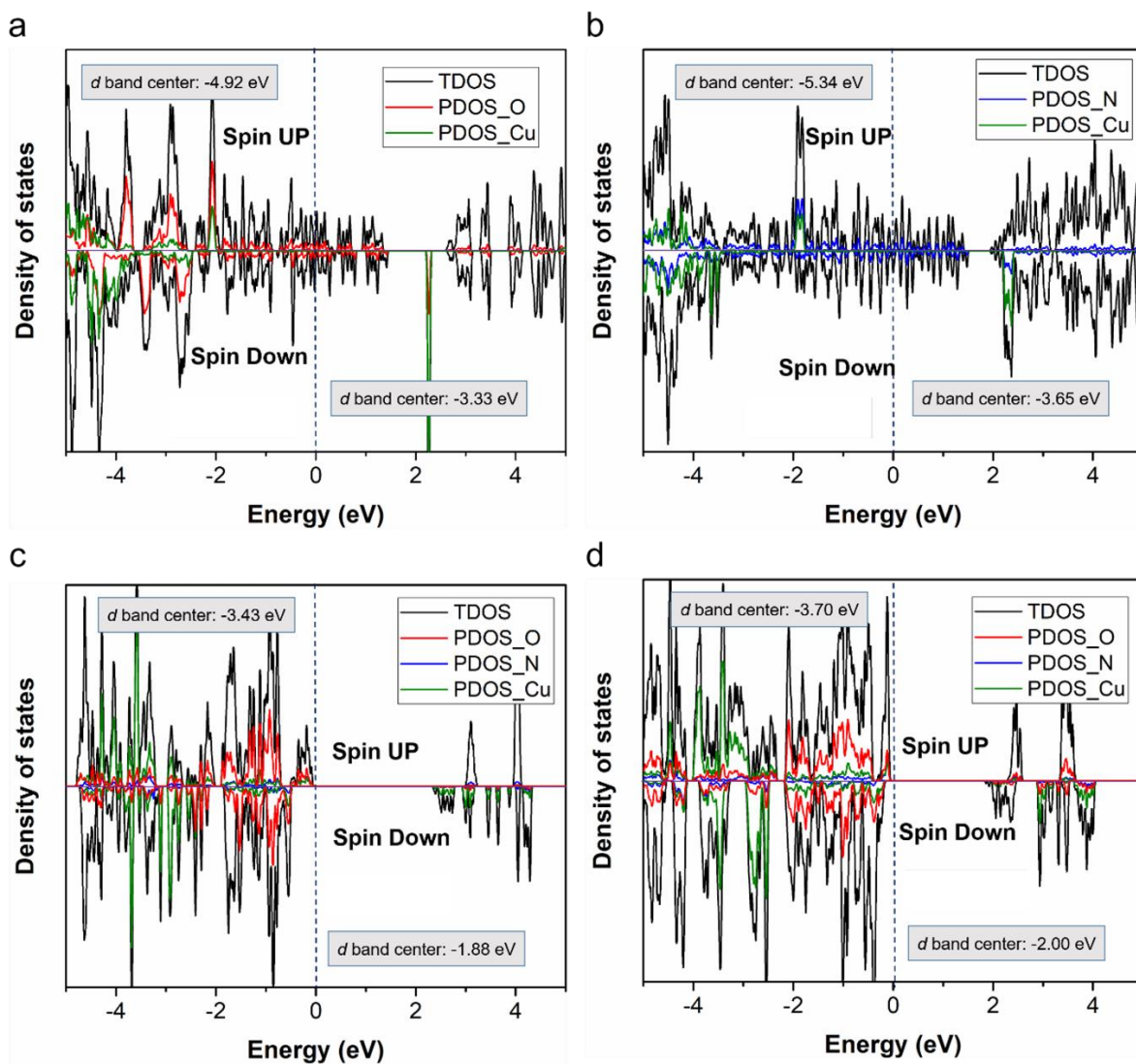


Figure S33. Total density of states (TDOS) and partial density of states (PDOS) of (a) Cu-HHTP, (b) Cu-HITP, (c) E-MCOF, and (d) Z-MCOF.

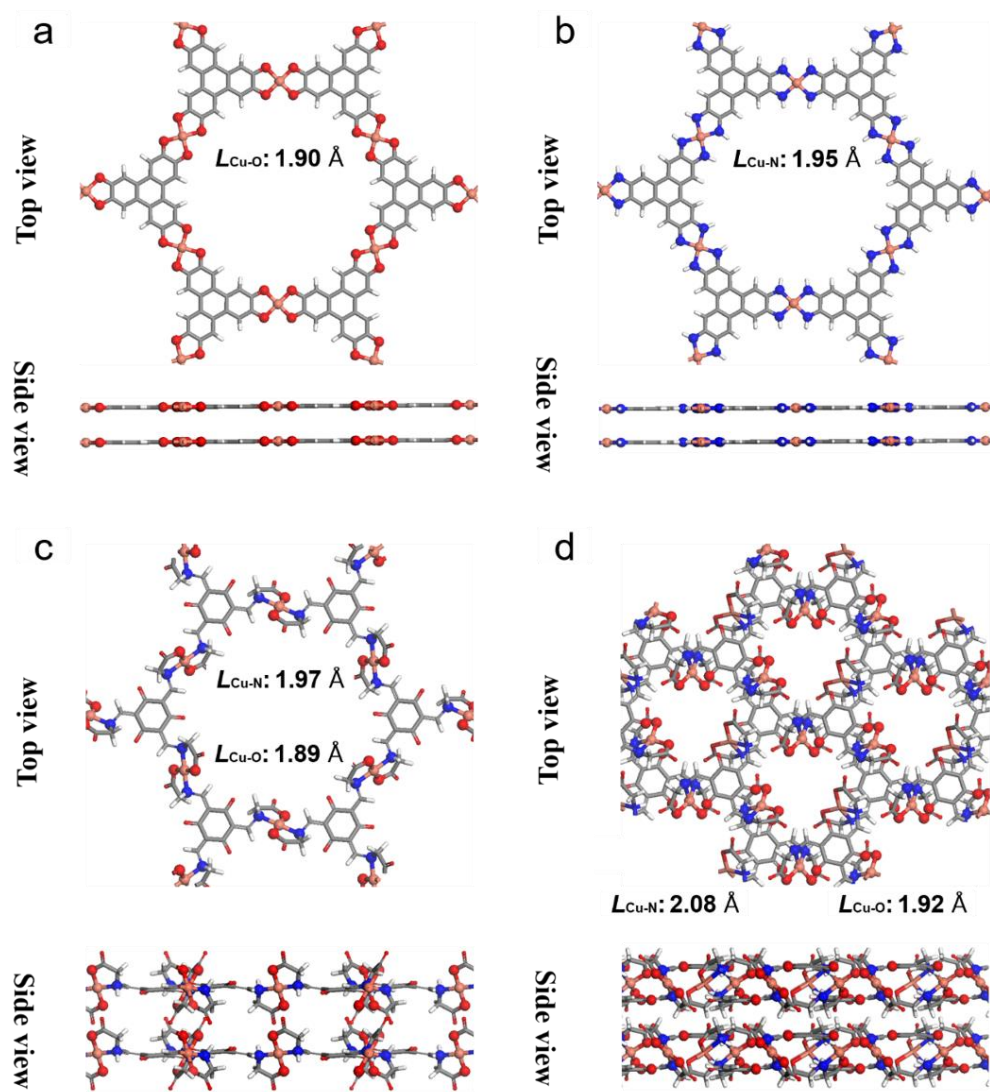


Figure S34. The optimized structures of (a) Cu-HHTP, (b) Cu-HITP, (c) Z-MCOF and (d) E-MCOF.

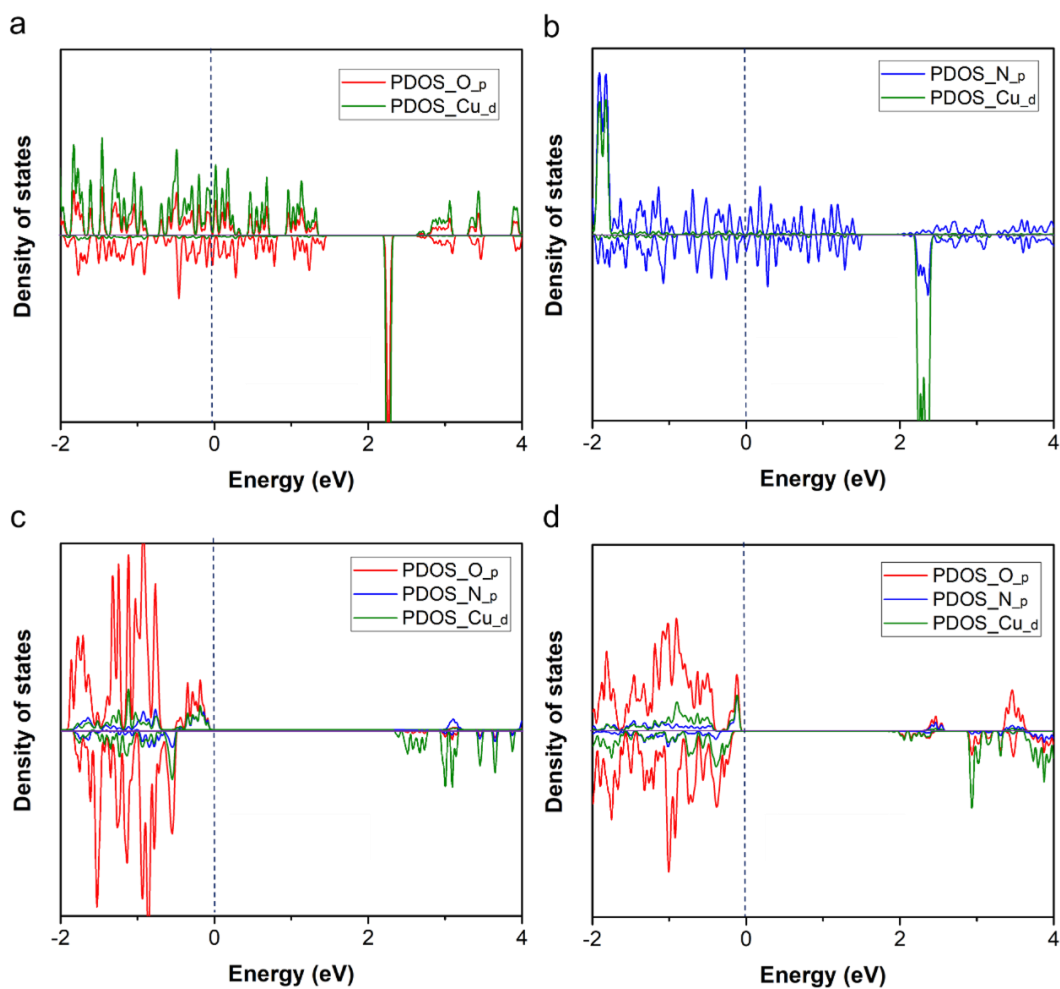


Figure S35. PDOS of p-orbitals and d-orbitals for (a) Cu-HHTP, (b) Cu-HITP, (c) E-MCOF, and (d) Z-MCOF.

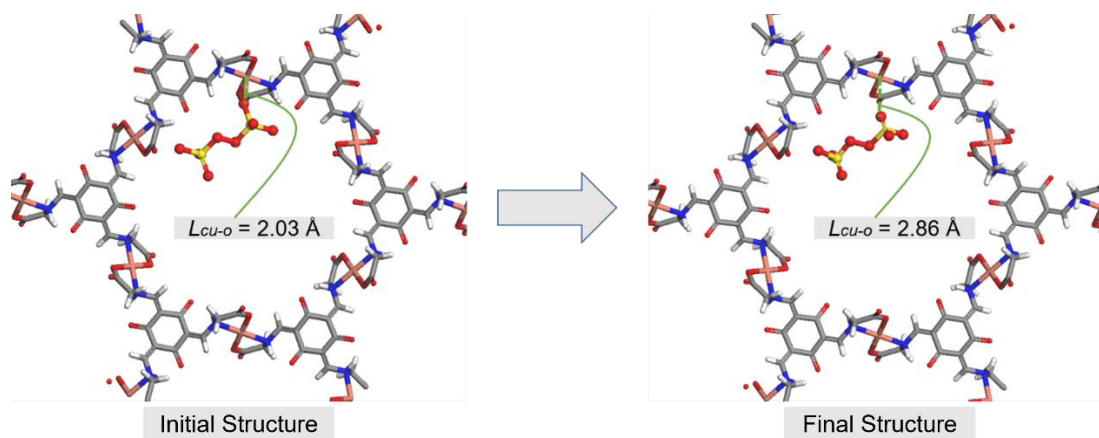


Figure S36. Geometry optimization of adsorption of $\text{K}_2\text{S}_2\text{O}_8$ on Z-MCOF bulk.

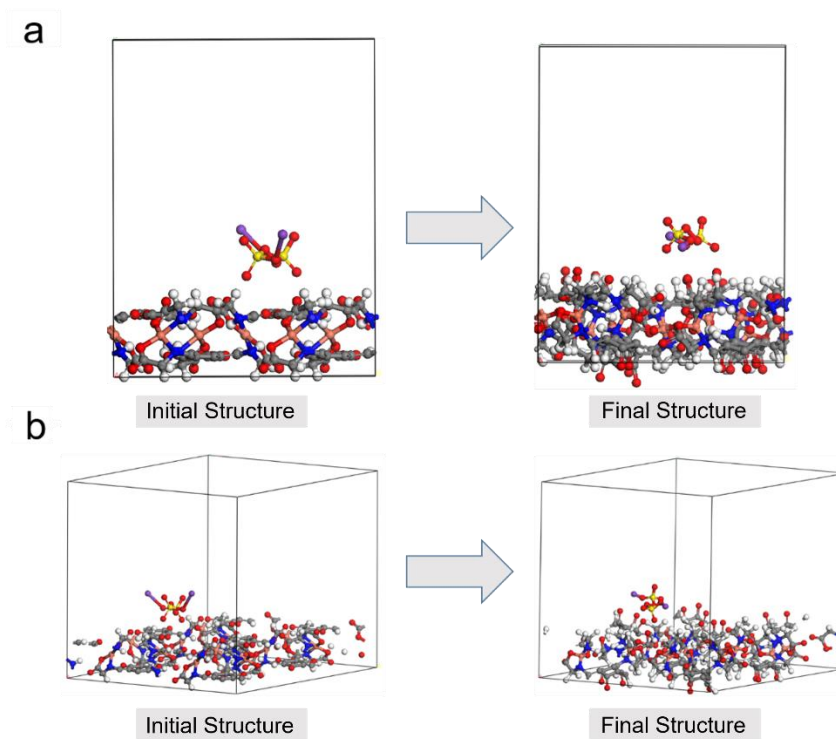


Figure S37. Geometry optimization of two possible adsorption modes of $\text{K}_2\text{S}_2\text{O}_8$ on E-MCOF surface.

Table S1. XPS element contents of MCOFs and Cu(Gly)₂.

Name	Atomic Concentration (%)			
	C1s	N1s	O1s	Cu2p3
E-MCOF	68.68	5.61	23.15	2.56
Z-MCOF	74.25	5.07	18.74	1.95
E-Cu(Gly) ₂	56.87	13.16	25.62	4.35
Z-Cu(Gly) ₂	54.18	10.83	30.11	4.88

Table S2. Unit cell parameters and fractional atomic coordinates for E-MCOF.

Formula: C ₃₀ H ₂₄ Cu ₃ N ₆ O ₁₈			
Space Group: <i>P</i> 3 2 1 (150)			
Crystal System: Trigonal			
Calculated Unit Cell: a = b = 12.749 Å, c = 5.882 Å, $\alpha = \beta = 90^\circ$, $\gamma = 120^\circ$			
Atom	<i>x</i>	<i>y</i>	<i>z</i>
H1	0.48156	0.56774	0.64235
H2	0.43226	0.91382	0.64235
H3	0.08618	0.51844	0.64235
H4	0.56774	0.48156	0.35765
H5	0.91382	0.43226	0.35765
H6	0.51844	0.08618	0.35765
H7	0.37217	0.56363	0.88793
H8	0.43637	0.80853	0.88793
H9	0.19147	0.62783	0.88793
H10	0.56363	0.37217	0.11207
H11	0.80853	0.43637	0.11207
H12	0.62783	0.19147	0.11207
H13	0.34863	0.42780	0.02042
H14	0.57220	0.92084	0.02042
H15	0.07916	0.65136	0.02042
H16	0.42780	0.34863	0.97957
H17	0.92084	0.57220	0.97957
H18	0.65136	0.07916	0.97957
H19	0.05378	0.41165	0.26163
H20	0.58835	0.64213	0.26163
H21	0.35787	0.94622	0.26163
H22	0.41165	0.05378	0.73837
H23	0.64213	0.58835	0.73837
H24	0.94622	0.35787	0.73837
N1	0.45389	0.48254	0.70890
N2	0.51746	0.97135	0.70890
N3	0.02865	0.54611	0.70890
N4	0.48254	0.45389	0.29111
N5	0.97135	0.51746	0.29111

N6	0.54611	0.02865	0.29111
C1	0.35290	0.47134	0.85957
C2	0.52866	0.88156	0.85957
C3	0.11844	0.64710	0.85957
C4	0.47134	0.35290	0.14043
C5	0.88156	0.52866	0.14043
C6	0.64710	0.11844	0.14043
C7	0.22431	0.39236	0.76557
C8	0.60764	0.83195	0.76557
C9	0.16805	0.77569	0.76557
C10	0.39236	0.22431	0.23443
C11	0.83195	0.60764	0.23443
C12	0.77569	0.16805	0.23443
C13	0.25443	0.72064	0.28422
C14	0.27936	0.53378	0.28422
C15	0.46622	0.74557	0.28422
C16	0.72064	0.25443	0.71578
C17	0.53378	0.27936	0.71578
C18	0.74557	0.46622	0.71578
C19	0.20041	0.58851	0.27872
C20	0.41149	0.61190	0.27872
C21	0.38810	0.79959	0.27872
C22	0.58851	0.20041	0.72128
C23	0.6119	0.41149	0.72128
C24	0.79959	0.38810	0.72128
C25	0.07754	0.50695	0.25994
C26	0.49305	0.57058	0.25994
C27	0.42942	0.92246	0.25994
C28	0.50695	0.07754	0.74006
C29	0.57058	0.49305	0.74006
C30	0.92246	0.42942	0.74006
O1	0.20147	0.29108	0.67676
O2	0.70892	0.91039	0.67676
O3	0.08961	0.79853	0.67676
O4	0.29108	0.20147	0.32324

O5	0.91039	0.70892	0.32324
O6	0.79853	0.08961	0.32324
O7	0.18619	0.76459	0.28633
O8	0.23541	0.42160	0.28633
O9	0.57840	0.81381	0.28633
O10	0.76459	0.18619	0.71367
O11	0.42160	0.23541	0.71367
O12	0.81381	0.57840	0.71367
O13	0.57110	0.72228	0.79334
O14	0.27772	0.84882	0.79334
O15	0.15118	0.42890	0.79334
O16	0.72228	0.57110	0.20666
O17	0.84882	0.27772	0.20666
O18	0.42890	0.15118	0.20666
Cu1	0.34639	0.34639	0.50000
Cu2	0.65361	0.00000	0.50000
Cu3	0.00000	0.65361	0.50000

Table S3. Unit cell parameters and fractional atomic coordinates for Z-MCOF.

Formula: C ₃₀ H ₃₆ Cu ₃ N ₆ O ₂₄			
Space Group: <i>P</i> - 3 (147)			
Crystal System: Trigonal			
Calculated Unit Cell: a = b = 18.269 Å, c = 7.217 Å, $\alpha = \beta = 90^\circ$, $\gamma = 120^\circ$			
Atom	<i>x</i>	<i>y</i>	<i>z</i>
C1	0.42980	0.03228	0.20408
C2	0.40347	0.94198	0.17211
C3	0.40377	0.55966	0.30283
C4	0.37238	0.61213	0.27731
C5	0.42543	0.70515	0.26150
C6	0.70515	0.27972	0.73850
C7	0.61213	0.23976	0.72269
C8	0.46149	0.40347	0.82789
C9	0.55966	0.15589	0.69717
C10	0.39752	0.42980	0.79592
C11	0.15589	0.59623	0.30283
C12	0.23976	0.62762	0.27731
C13	0.27972	0.57457	0.26150
C14	0.94198	0.53851	0.82789
C15	0.03228	0.60248	0.79592
C16	0.44034	0.84411	0.30283
C17	0.96772	0.39752	0.20408
C18	0.05802	0.46149	0.17211
C19	0.38787	0.76024	0.27731
C20	0.29485	0.72028	0.26150
C21	0.59653	0.05802	0.82789
C22	0.72028	0.42543	0.73850
C23	0.76024	0.37238	0.72269
C24	0.57020	0.96772	0.79592
C25	0.84411	0.40377	0.69717
C26	0.59623	0.44034	0.69717
C27	0.62762	0.38787	0.72269
C28	0.57457	0.29485	0.73850
C29	0.53851	0.59653	0.17211

C30	0.60248	0.57020	0.20408
N1	0.41455	0.90500	0.34378
N2	0.90500	0.49045	0.65622
N3	0.50955	0.41455	0.65622
N4	0.58545	0.09500	0.65622
N5	0.09500	0.50955	0.34378
N6	0.49045	0.58545	0.34378
O1	0.43410	0.41517	0.32271
O2	0.92928	0.34232	0.08958
O3	0.58483	0.01893	0.32271
O4	0.47495	0.07315	0.36281
O5	0.58696	0.92928	0.91042
O6	0.56590	0.58483	0.67729
O7	0.73709	0.23684	0.77169
O8	0.23684	0.49976	0.22831
O9	0.34232	0.41304	0.91042
O10	0.59820	0.52505	0.36281
O11	0.52505	0.92685	0.63719
O12	0.41304	0.07072	0.08958
O13	0.50024	0.73709	0.22831
O14	0.07072	0.65768	0.91042
O15	0.07315	0.59820	0.63719
O16	0.76316	0.50024	0.77169
O17	0.01893	0.43410	0.67729
O18	0.49976	0.26291	0.77169
O19	0.98107	0.56590	0.32271
O20	0.40180	0.47495	0.63719
O21	0.65768	0.58696	0.08958
O22	0.26291	0.76316	0.22831
O23	0.41517	0.98107	0.67729
O24	0.92685	0.40180	0.36281
H1	0.12378	0.48204	0.41818
H2	0.36955	0.98431	0.61140
H3	0.98431	0.61476	0.38860
H4	0.92216	0.53287	0.27934

H5	0.09132	0.42789	0.12857
H6	0.86738	0.36022	0.69860
H7	0.48204	0.35825	0.58182
H8	0.06161	0.50277	0.05552
H9	0.49723	0.55885	0.05552
H10	0.63978	0.50716	0.69860
H11	0.57211	0.66343	0.12857
H12	0.51796	0.64175	0.41818
H13	0.42789	0.33657	0.87143
H14	0.07784	0.46713	0.72066
H15	0.01569	0.38524	0.61140
H16	0.36022	0.49284	0.30140
H17	0.50277	0.44115	0.94448
H18	0.44115	0.93839	0.05552
H19	0.63045	0.01569	0.38860
H20	0.61071	0.07784	0.27934
H21	0.87622	0.51796	0.58182
H22	0.90868	0.57211	0.87143
H23	0.13262	0.63978	0.30140
H24	0.61476	0.63045	0.61140
H25	0.55885	0.06161	0.94448
H26	0.50716	0.86738	0.30140
H27	0.66343	0.09132	0.87143
H28	0.64175	0.12378	0.58182
H29	0.93839	0.49723	0.94448
H30	0.46713	0.38929	0.27934
H31	0.38524	0.36955	0.38860
H32	0.53287	0.61071	0.72066
H33	0.49284	0.13262	0.69860
H34	0.33657	0.90868	0.12857
H35	0.35825	0.87622	0.41818
H36	0.38929	0.92216	0.72066
Cu1	0.50000	0.50000	0.50000
Cu2	0.50000	0.00000	0.50000
Cu3	0.00000	0.50000	0.50000

Table S4. The calculated lattice parameters, and energies.

Formula	Lattice Parameters						Energy (eV)
	$a/\text{\AA}$	$b/\text{\AA}$	$c/\text{\AA}$	$\alpha/^\circ$	$\beta/^\circ$	$\gamma/^\circ$	
Cu-HHTP	21.75	21.75	3.31	90	90	120	-459.06
Cu-HITP	22.18	22.18	3.33	90	90	120	-523.81
<i>Z-MCOF</i>	18.87	18.87	5.21	90	90	120	-539.86
<i>E-MCOF</i>	11.89	11.89	6.82	90	90	120	-529.27
K ₂ S ₂ O ₈ @Z-MCOF bulk	37.73	18.87	10.43	90	90	120	-2080.51
Z-MCOF slab	33.47	19.33	21.97	90	90	90	-1076.28
K ₂ S ₂ O ₈ @Z-MCOF slab	33.47	19.33	21.97	90	90	90	-1141.95
							($E_{\text{ads}} = -1.33$)
SO ₄ @Z-MCOF	33.47	19.33	21.97	90	90	90	-1105.14
K ₂ S ₂ O ₈ @E-MCOF	23.77	23.77	26.51	90	90	120	—
K ₂ S ₂ O ₈	33.47	19.33	21.97	90	90	90	64.34 eV

Table S5. Unit cell parameters and fractional atomic coordinates for Z-MCOF₀.

Formula: C ₃₀ H ₂₄ Cu ₃ N ₆ O ₁₈			
Space Group: <i>P</i> 6/ <i>m</i> (175)			
Crystal System: Hexagonal			
Calculated Unit Cell: <i>a</i> = <i>b</i> = 20.613 Å, <i>c</i> = 4.041 Å, $\alpha = \beta = 90^\circ$, $\gamma = 120^\circ$			
Atom	<i>x</i>	<i>y</i>	<i>z</i>
C1	0.58842	0.42343	0.00000
C2	0.65873	0.10692	0.00000
C3	0.64492	0.02909	0.00000
C4	0.25195	0.62638	0.00000
C5	0.42343	0.83501	0.00000
C6	0.16499	0.58842	0.00000
C7	0.37443	0.74805	0.00000
C8	0.29468	0.70718	0.00000
C9	0.97091	0.61583	0.00000
C10	0.44819	0.34127	0.00000
C11	0.38417	0.35508	0.00000
C12	0.89308	0.55181	0.00000
C13	0.41250	0.70532	0.00000
C14	0.61583	0.64492	0.00000
C15	0.55181	0.65873	0.00000
C16	0.62638	0.37443	0.00000
C17	0.58750	0.29468	0.00000
C18	0.70532	0.29282	0.00000
C19	0.62557	0.25195	0.00000
C20	0.37362	0.62557	0.00000
C21	0.57657	0.16499	0.00000
C22	0.41158	0.57657	0.00000
C23	0.10692	0.44819	0.00000
C24	0.35508	0.97091	0.00000
C25	0.34127	0.89308	0.00000
C26	0.02909	0.38417	0.00000
C27	0.83501	0.41158	0.00000
C28	0.70718	0.41250	0.00000
C29	0.74805	0.37362	0.00000

C30	0.29282	0.58750	0.00000
N1	0.58829	0.10738	0.00000
N2	0.41171	0.89262	0.00000
N3	0.10738	0.51909	0.00000
N4	0.48091	0.58829	0.00000
N5	0.89262	0.48091	0.00000
N6	0.51909	0.41171	0.00000
O1	0.03069	0.60326	0.00000
O2	0.39674	0.42743	0.00000
O3	0.32006	0.30246	0.00000
O4	0.98240	0.67994	0.00000
O5	0.26037	0.75037	0.00000
O6	0.01760	0.32006	0.00000
O7	0.49000	0.73963	0.00000
O8	0.24963	0.51000	0.00000
O9	0.96931	0.39674	0.00000
O10	0.75037	0.49000	0.00000
O11	0.30246	0.98240	0.00000
O12	0.57257	0.96931	0.00000
O13	0.42743	0.03069	0.00000
O14	0.73963	0.24963	0.00000
O15	0.67994	0.69754	0.00000
O16	0.51000	0.26037	0.00000
O17	0.60326	0.57257	0.00000
O18	0.69754	0.01760	0.00000
H1	0.37383	0.51983	0.00000
H2	0.52756	0.79443	0.00000
H3	0.51983	0.14600	0.00000
H4	0.20557	0.73313	0.00000
H5	0.26687	0.47244	0.00000
H6	0.47244	0.20557	0.00000
H7	0.79443	0.26687	0.00000
H8	0.55716	0.69040	-0.22853
H9	0.62617	0.48017	0.00000
H10	0.85400	0.37383	0.00000

H11	0.30960	0.86676	0.22853
H12	0.55716	0.69040	0.22853
H13	0.86676	0.55716	0.22853
H14	0.14600	0.62617	0.00000
H15	0.69040	0.13324	0.22853
H16	0.44284	0.30960	0.22853
H17	0.86676	0.55716	-0.22853
H18	0.69040	0.13324	-0.22853
H19	0.48017	0.85400	0.00000
H20	0.44284	0.30960	-0.22853
H21	0.13324	0.44284	-0.22853
H22	0.30960	0.86676	-0.22853
H23	0.13324	0.44284	0.22853
H24	0.73313	0.52756	0.00000
Cu1	0.50000	0.00000	0.00000
Cu2	0.50000	0.50000	0.00000
Cu3	0.00000	0.50000	0.00000

Table S6. Unit cell parameters and fractional atomic coordinates for Z-MCOF₆₀.

Formula: C ₃₀ H ₂₄ Cu ₃ N ₆ O ₁₈			
Space Group: <i>P</i> - 3 (147)			
Crystal System: Trigonal			
Calculated Unit Cell: a = b = 19.327 Å, c = 6.093 Å, $\alpha = \beta = 90^\circ$, $\gamma = 120^\circ$			
Atom	<i>x</i>	<i>y</i>	<i>z</i>
C1	0.42377	0.98184	0.10894
C2	0.57623	0.01816	0.89106
C3	0.01816	0.44192	0.10894
C4	0.98184	0.55808	0.89106
C5	0.55808	0.57623	0.10894
C6	0.44192	0.42377	0.89106
C7	0.40903	0.89748	0.15069
C8	0.59097	0.10252	0.84931
C9	0.10252	0.51155	0.15069
C10	0.89748	0.48845	0.84931
C11	0.48845	0.59097	0.15069
C12	0.51155	0.40903	0.84931
C13	0.69731	0.27643	0.45656
C14	0.30268	0.72357	0.54344
C15	0.72357	0.42089	0.45656
C16	0.27643	0.57911	0.54344
C17	0.57911	0.30268	0.45656
C18	0.42089	0.69731	0.54344
C19	0.61101	0.24762	0.47988
C20	0.38899	0.75238	0.52012
C21	0.75238	0.36338	0.47988
C22	0.24762	0.63661	0.52012
C23	0.63661	0.38899	0.47988
C24	0.36338	0.61101	0.52012
C25	0.56157	0.17404	0.57133
C26	0.43843	0.82596	0.42867
C27	0.82596	0.38753	0.57133
C28	0.17404	0.61247	0.42867
C29	0.61247	0.43843	0.57133

C30	0.38753	0.56157	0.42867
N1	0.41598	0.88316	0.38878
N2	0.58402	0.11684	0.61122
N3	0.11684	0.53282	0.38878
N4	0.88316	0.46718	0.61122
N5	0.46718	0.58402	0.38878
N6	0.53282	0.41598	0.61122
O1	0.40679	-0.00167	0.92788
O2	0.59321	0.00167	0.07212
O3	0.00167	0.40846	0.92788
O4	-0.00167	0.59154	0.07212
O5	0.59154	0.59321	0.92788
O6	0.40846	0.40679	0.07212
O7	0.54189	0.96657	0.73215
O8	0.45812	0.03343	0.26785
O9	0.03343	0.57532	0.73215
O10	0.96657	0.42468	0.26785
O11	0.42468	0.45812	0.73215
O12	0.57532	0.54189	0.26785
O13	0.72319	0.22923	0.43407
O14	0.27681	0.77077	0.56593
O15	0.77077	0.49396	0.43407
O16	0.22923	0.50603	0.56593
O17	0.50603	0.27681	0.43407
O18	0.49396	0.72319	0.56593
H1	0.45454	0.89126	0.05849
H2	0.54546	0.10874	0.94151
H3	0.10874	0.56328	0.05849
H4	0.89126	0.43672	0.94151
H5	0.43672	0.54546	0.05849
H6	0.56328	0.45454	0.94151
H7	0.5024	0.15937	0.63102
H8	0.4976	0.84063	0.36898
H9	0.84063	0.34303	0.63102
H10	0.15937	0.65697	0.36898

H11	0.65697	0.49760	0.63102
H12	0.34303	0.50240	0.36898
H13	0.34968	0.85277	0.08994
H14	0.65032	0.14723	0.91006
H15	0.14723	0.49691	0.08994
H16	0.85277	0.50309	0.91006
H17	0.50309	0.65032	0.08994
H18	0.49691	0.34968	0.91006
H19	0.36224	0.86557	0.46937
H20	0.63776	0.13443	0.53063
H21	0.13443	0.49666	0.46937
H22	0.86557	0.50334	0.53063
H23	0.50334	0.63776	0.46937
H24	0.49666	0.36224	0.53063
Cu1	0.50000	0.50000	0.50000
Cu2	0.50000	0.00000	0.50000
Cu3	0.00000	0.50000	0.50000

Table S7. Unit cell parameters and fractional atomic coordinates for Z-MCOF₁₀₅.

Formula: C ₃₀ H ₂₄ Cu ₃ N ₆ O ₁₈			
Space Group: <i>P</i> - 3 (147)			
Crystal System: Trigonal			
Calculated Unit Cell: a = b = 16.100 Å, c = 6.093 Å, $\alpha = \beta = 90^\circ$, $\gamma = 120^\circ$			
Atom	<i>x</i>	<i>y</i>	<i>z</i>
C1	0.60727	0.03701	-0.36384
C2	0.59044	0.94039	-0.30184
C3	0.29774	0.47539	-0.02219
C4	0.31680	0.56663	-0.05914
C5	0.41515	0.64996	-0.07920
C6	0.64996	0.23481	0.07920
C7	0.56663	0.24983	0.05914
C8	0.65005	0.59044	0.30184
C9	0.47539	0.17765	0.02219
C10	0.57026	0.60727	0.36384
C11	0.17765	0.70226	-0.02219
C12	0.24983	0.68320	-0.05914
C13	0.23481	0.58485	-0.07920
C14	0.96299	0.57026	-0.36384
C15	0.05961	0.65005	-0.30184
C16	0.94039	0.34995	0.30184
C17	0.03701	0.42974	0.36384
C18	0.52461	0.82235	-0.02219
C19	0.76519	0.41515	0.07920
C20	0.75017	0.31680	0.05914
C21	0.43337	0.75017	-0.05914
C22	0.82235	0.29774	0.02219
C23	0.35004	0.76519	-0.07920
C24	0.40956	0.05961	0.30184
C25	0.39273	0.96299	0.36384
C26	0.70226	0.52461	0.02219
C27	0.42974	0.39273	-0.36384
C28	0.34995	0.40956	-0.30184
C29	0.68320	0.43337	0.05914

C30	0.58485	0.35004	0.07920
N1	0.07671	0.63934	-0.07037
N2	0.63934	0.56263	0.07037
N3	0.56263	0.92329	-0.07037
N4	0.43737	0.07671	0.07037
N5	0.92329	0.36066	0.07037
N6	0.36066	0.43737	-0.07037
O1	0.15510	0.51829	-0.11973
O2	0.92753	0.48208	-0.26027
O3	0.91539	0.58197	-0.50299
O4	0.63681	0.15510	0.11973
O5	0.07247	0.51792	0.26027
O6	0.58197	0.66658	0.50299
O7	0.51792	0.44545	-0.26027
O8	0.44545	0.92753	0.26027
O9	0.66658	0.08461	-0.50299
O10	0.48171	0.63681	-0.11973
O11	0.36319	0.84490	-0.11973
O12	0.84490	0.48171	0.11973
O13	0.55455	0.07247	-0.26027
O14	0.33342	0.91539	0.50299
O15	0.51829	0.36319	0.11973
O16	0.41803	0.33342	-0.50299
O17	0.08461	0.41803	0.50299
O18	0.48208	0.55455	0.26027
H1	0.66619	0.62597	-0.02123
H2	0.04022	0.66619	0.02123
H3	0.27942	0.34369	-0.33108
H4	0.77655	0.57838	-0.00858
H5	0.35496	0.46809	-0.40821
H6	0.33381	0.37403	0.02123
H7	0.11313	0.64504	-0.40821
H8	0.80183	0.22345	-0.00858
H9	0.93573	0.27942	0.33108
H10	0.64504	0.53191	0.40821

H11	0.22345	0.42162	0.00858
H12	0.06427	0.72058	-0.33108
H13	0.72058	0.65631	0.33108
H14	0.42162	0.19817	-0.00858
H15	0.88687	0.35496	0.40821
H16	0.34369	0.06427	0.33108
H17	0.57838	0.80183	0.00858
H18	0.46809	0.11313	0.40821
H19	0.37403	0.04022	-0.02123
H20	0.65631	0.93573	-0.33108
H21	0.53191	0.88687	-0.40821
H22	0.62597	0.95978	0.02123
H23	0.95978	0.33381	-0.02123
H24	0.19817	0.77655	0.00858
Cu1	0.00000	0.50000	0.00000
Cu2	0.50000	0.50000	0.00000
Cu3	0.50000	0.00000	0.00000

Table S8. Unit cell parameters and fractional atomic coordinates for Cu-HHTP.

Formula: C ₃₆ H ₁₂ Cu ₃ O ₁₂			
Space Group: <i>P</i> 6/ <i>m</i> <i>m</i> <i>m</i> (191)			
Crystal System: Hexagonal			
Calculated Unit Cell: <i>a</i> = <i>b</i> = 21.797 Å, <i>c</i> = 3.441 Å, $\alpha = \beta = 90^\circ$, $\gamma = 120^\circ$			
Atom	<i>x</i>	<i>y</i>	<i>z</i>
C1	0.46388	0.66587	0.00000
C2	0.53612	0.33413	0.00000
C3	0.33413	0.79802	0.00000
C4	0.66587	0.20198	0.00000
C5	0.20198	0.53612	0.00000
C6	0.79802	0.46388	0.00000
C7	0.66587	0.46388	0.00000
C8	0.33413	0.53612	0.00000
C9	0.79802	0.33413	0.00000
C10	0.20198	0.66587	0.00000
C11	0.53612	0.20198	0.00000
C12	0.46388	0.79802	0.00000
C13	0.39792	0.53493	0.00000
C14	0.60208	0.46507	0.00000
C15	0.46507	0.86299	0.00000
C16	0.53493	0.13701	0.00000
C17	0.13701	0.60208	0.00000
C18	0.86299	0.39792	0.00000
C19	0.53493	0.39792	0.00000
C20	0.46507	0.60208	0.00000
C21	0.86299	0.46507	0.00000
C22	0.13701	0.53493	0.00000
C23	0.60208	0.13701	0.00000
C24	0.39792	0.86299	0.00000
C25	0.33321	0.59981	0.00000
C26	0.66679	0.40019	0.00000
C27	0.40019	0.73339	0.00000
C28	0.59981	0.26660	0.00000
C29	0.26660	0.66679	0.00000

C30	0.73339	0.33321	0.00000
C31	0.59981	0.33321	0.00000
C32	0.40019	0.66679	0.00000
C33	0.73339	0.40019	0.00000
C34	0.26660	0.59981	0.00000
C35	0.66679	0.26660	0.00000
C36	0.33321	0.73339	0.00000
O1	0.40177	0.47752	0.00000
O2	0.59823	0.52248	0.00000
O3	0.52248	0.92425	0.00000
O4	0.47752	0.07575	0.00000
O5	0.07575	0.59823	0.00000
O6	0.92425	0.40177	0.00000
O7	0.47752	0.40177	0.00000
O8	0.52248	0.59823	0.00000
O9	0.92425	0.52248	0.00000
O10	0.07575	0.47752	0.00000
O11	0.59823	0.07575	0.00000
O12	0.40177	0.92425	0.00000
H1	0.51508	0.71434	0.00000
H2	0.48492	0.28566	0.00000
H3	0.28566	0.80073	0.00000
H4	0.71434	0.19927	0.00000
H5	0.19927	0.48492	0.00000
H6	0.80073	0.51508	0.00000
H7	0.71434	0.51508	0.00000
H8	0.28566	0.48492	0.00000
H9	0.80073	0.28566	0.00000
H10	0.19927	0.71434	0.00000
H11	0.48492	0.19927	0.00000
H12	0.51508	0.80073	0.00000
Cu1	0.00000	0.50000	0.00000
Cu2	0.50000	0.50000	0.00000
Cu3	0.50000	0.00000	0.00000

Table S9. Unit cell parameters and fractional atomic coordinates for Cu-HITP.

Formula: C ₃₆ H ₂₄ Cu ₃ N ₁₂			
Space Group: <i>P</i> 6/ <i>m</i> <i>m</i> <i>m</i> (191)			
Crystal System: Hexagonal			
Calculated Unit Cell: <i>a</i> = <i>b</i> = 22.227 Å, <i>c</i> = 3.424 Å, $\alpha = \beta = 90^\circ$, $\gamma = 120^\circ$			
Atom	<i>x</i>	<i>y</i>	<i>z</i>
C1	0.53922	0.20437	0.00000
C2	0.46078	0.79563	0.00000
C3	0.79563	0.33485	0.00000
C4	0.20437	0.66515	0.00000
C5	0.66515	0.46078	0.00000
C6	0.33485	0.53922	0.00000
C7	0.20437	0.53922	0.00000
C8	0.79563	0.46078	0.00000
C9	0.33485	0.79563	0.00000
C10	0.66515	0.20437	0.00000
C11	0.46078	0.66515	0.00000
C12	0.53922	0.33485	0.00000
C13	0.60292	0.14000	0.00000
C14	0.39708	0.86000	0.00000
C15	0.86000	0.46291	0.00000
C16	0.14000	0.53709	0.00000
C17	0.53709	0.39708	0.00000
C18	0.46291	0.60292	0.00000
C19	0.14000	0.60292	0.00000
C20	0.86000	0.39708	0.00000
C21	0.46291	0.86000	0.00000
C22	0.53709	0.14000	0.00000
C23	0.39708	0.53709	0.00000
C24	0.60292	0.46291	0.00000
C25	0.66663	0.26788	0.00000
C26	0.33337	0.73212	0.00000
C27	0.73212	0.39875	0.00000
C28	0.26788	0.60125	0.00000
C29	0.60125	0.33337	0.00000

C30	0.39875	0.66663	0.00000
C31	0.26788	0.66663	0.00000
C32	0.73212	0.33337	0.00000
C33	0.39875	0.73212	0.00000
C34	0.60125	0.26788	0.00000
C35	0.33337	0.60125	0.00000
C36	0.66663	0.39875	0.00000
N1	0.59567	0.07675	0.00000
N2	0.40433	0.92325	0.00000
N3	0.92325	0.51892	0.00000
N4	0.07675	0.48108	0.00000
N5	0.48108	0.40433	0.00000
N6	0.51892	0.59567	0.00000
N7	0.07675	0.59567	0.00000
N8	0.92325	0.40433	0.00000
N9	0.51892	0.92325	0.00000
N10	0.48108	0.07675	0.00000
N11	0.40433	0.48108	0.00000
N12	0.59567	0.51892	0.00000
H1	0.48947	0.20290	0.00000
H2	0.51053	0.79710	0.00000
H3	0.79710	0.28657	0.00000
H4	0.20290	0.71343	0.00000
H5	0.71343	0.51053	0.00000
H6	0.28657	0.48947	0.00000
H7	0.20290	0.48947	0.00000
H8	0.79710	0.51053	0.00000
H9	0.28657	0.79710	0.00000
H10	0.71343	0.20290	0.00000
H11	0.51053	0.71343	0.00000
H12	0.48947	0.28657	0.00000
H13	0.64079	0.07523	0.00000
H14	0.35921	0.92477	0.00000
H15	0.92477	0.56556	0.00000
H16	0.07523	0.43444	0.00000

H17	0.43444	0.35921	0.00000
H18	0.56556	0.64079	0.00000
H19	0.07523	0.64079	0.00000
H20	0.92477	0.35921	0.00000
H21	0.56556	0.92477	0.00000
H22	0.43444	0.07523	0.00000
H23	0.35921	0.43444	0.00000
H24	0.64079	0.56556	0.00000
Cu1	0.00000	0.50000	0.00000
Cu2	0.50000	0.50000	0.00000
Cu3	0.50000	0.00000	0.00000

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